

Cover Letter

May 28, 2026

Dr. Chris Paton
Editor in Chief
BMJ Digital Health & AI

Dear Dr. Paton,

We are pleased to resubmit our manuscript, “**CLINES: Clinical LLM-based Information Extraction and Structuring Agent**” (manuscript ID bmjdh-2026-000027), revised in response to the Major Revision decision letter dated [date]. We thank the editorial office and the five reviewers for the depth and rigor of the feedback, which has materially strengthened the manuscript.

The revised manuscript incorporates substantial additions to address the reviewers’ major concerns. Specifically, we have:

- conducted a structured **inter-annotator agreement (IAA)** study on six cross-annotated notes (three per dataset for 4CE and CORAL), reporting an F1-based IAA of 0.837 (Cohen’s $\kappa = 0.597$, PABAK = 0.609) following pre-specified structural-disambiguation guidelines, and explicitly reframed MIMIC-III F1 as agreement-with-annotator (addressing R1.1, R2.2, R5.4.2);
- added **bootstrap 95% confidence intervals** and permutation tests to every primary F1 comparison (R1.2), and a document-level resampling stability analysis (R1.3);
- completed a **seven-class hallucination taxonomy** with stratified manual judging of 700 candidate false positives, finding that 54.3% of apparent false positives are not true model errors but rather annotation gaps, with each true-hallucination category accounting for $\leq 17\%$ of cases (R1.9, R2.3, R4.3, R5.3.2);
- added **cost, latency, and computational efficiency** reporting per dataset and per model, including total LLM invocations per note and Azure OpenAI list-price USD per note (R1.5, R4.4);
- expanded the baseline cohort with **o3-mini single-prompt**, **GPT-4o chain-of-thought single-prompt**, and two off-the-shelf clinical deep-learning encoders (**BERT-base clinical NER** and **GatorTron-base**), closing the previous baseline asymmetry (R1.7, R3.8, R5.4.1);
- performed a **component-wise ablation** of the four pipeline stages, isolating the contribution of semantic chunking, SapBERT normalization, the date module, and cross-chunk reconciliation (R1.8, R3.5);
- replaced the STROBE reporting checklist with **MI-CLAIM** and **TRIPOD+AI** checklists in the supplementary materials (R1.4, R5.3.3);
- reframed all instances of “agentic” and “zero-shot” to more precise language (R1.6, R2.1), tempered scalability and interoperability claims with explicit discussion of FP8 precision for

Llama-3.1-405B, GDPR considerations, and UMLS-to-SNOMED/LOINC/RxNorm/OMOP-CDM mapping requirements (R5.4.3, R5.4.4);

- added the BMJ DH&AI “Objective / Methods and Analysis / Results / Conclusion” abstract structure plus the required summary box (R1.18, R5.2.1, R5.2.2);
- clarified annotator credentialing (all annotators completed CITI Human Subjects Research training) and explicitly stated that LLM inference used the **HMS HIPAA-compliant Azure OpenAI tenant** under a Business Associate Agreement with Microsoft, with no clinical data transmitted to OpenAI directly (R1.15);
- added a Patient and Public Involvement statement (R1.20), replaced the vague ethics statement with an accurate one—under the U.S. Common Rule (45 CFR 46.104) the secondary analysis of fully de-identified data does not constitute human-subjects research and required no IRB approval at the authors’ institution (R1.21)—and corrected the i2b2 2010 citation to Uzuner et al. 2011 (R1.10);
- incorporated and discussed the three contemporary comparators highlighted by Reviewer 5 (John Snow Labs Healthcare NLP, CLEAR, and Ntinopoulos et al. 2025), with reasoned positioning where direct empirical comparison was not feasible (R5.4.5).

A detailed point-by-point response to each reviewer comment accompanies this letter. All textual changes in the revised manuscript are marked in [blue](#); the supplementary materials are also updated accordingly. We are submitting:

1. This cover letter
2. Point-by-point response to reviewers
3. Revised manuscript with all changes marked (blue)
4. Clean revised manuscript (no markup, for typesetting)
5. Original (submitted) manuscript for reference
6. Updated supplementary materials including MI-CLAIM and TRIPOD+AI checklists

We confirm that there are **no changes to author conflicts of interest** from the initial submission. No new funding sources were involved in this revision. All authors have read and approved the revised manuscript and accept responsibility for its accuracy and integrity. The work has not been submitted elsewhere and remains exclusively under consideration at *BMJ Digital Health & AI*.

We hope the revised version addresses the reviewers’ concerns and would be grateful for the opportunity to have it considered for publication. Please do not hesitate to contact the corresponding authors if any further clarification or material is required.

Sincerely,

Zongxin Yang, MS (first author)
Isaac Kohane, MD, PhD (co-corresponding author)
Tianxi Cai, ScD (co-corresponding author)
on behalf of all co-authors

Response to Reviewers

CLINES: Clinical LLM-based Information Extraction and Structuring Agent

BMJ Digital Health & AI — Manuscript bmjdh-2026-000027

May 26, 2026

Overview

We thank the Editor and the five reviewers for their detailed and constructive feedback. The revised manuscript incorporates substantial additions in response to the major concerns raised, including:

- A structured **inter-annotator agreement (IAA)** study with cross-annotation on three notes per dataset (4CE, CORAL), reporting F1-based IAA, Cohen’s κ , and PABAK (Methods (Inter-annotator agreement); Figure 4a).
- **Bootstrap 95% confidence intervals** and permutation tests on all primary F1 comparisons (the Results; Figure 5a).
- A **seven-class hallucination taxonomy** with stratified manual judging of 700 candidate false positives, extrapolated population-level rates, and full case studies (Methods; the Results; Figure 4c-d).
- **Cost, latency, and computational efficiency** reporting per dataset, per model, and per LLM-invocation type (the Results; Figure 5b).
- **Expanded baselines:** o3-mini single-prompt and GPT-4o chain-of-thought conditions (addressing the previous baseline asymmetry), plus two off-the-shelf clinical deep-learning encoders (BERT-base clinical NER and GatorTron-base) for direct LLM-vs-DL comparison (Results; Figure 5d).
- **Component-wise ablation** of the four pipeline stages (SapBERT normalization, semantic chunking, date module, cross-chunk reconciliation; Figure 5c).
- **MI-CLAIM and TRIPOD+AI** reporting checklists in place of STROBE (supplementary materials).
- Multiple smaller clarifications regarding annotation protocol, data governance, computational requirements, and downstream interoperability.

All changes in the revised manuscript are marked in [blue](#). Cross-references in this document point to the revised manuscript page/section numbers.

1 Reviewer 1

Major concerns

R1.1 — Missing Inter-Annotator Agreement

“The absence of inter-annotator agreement (IAA) reporting is the most significant methodological weakness... I would strongly encourage the authors to either report IAA for existing annotations or double-annotate a representative subset and report agreement statistics (e.g., Cohen’s kappa, F1-based IAA) in a revised manuscript.”

We thank the reviewer for this critical observation. We have conducted a structured cross-annotation study on three notes per dataset (4CE $n = 3$; CORAL $n = 3$; total 2,657 candidate-mention rows). For each note, a second annotator, blinded to the original assignment, independently re-annotated the AI-suggested draft. We report F1-based IAA as the primary metric (Hripcsak & Rothschild, JAMIA 2005) because the class distribution is highly skewed ($\approx 75\%$ keep rate), conditions under which Cohen’s κ suffers from Cohen’s paradox (Feinstein & Cicchetti, JCE 1990); Cohen’s κ and PABAK (Byrt et al., JCE 1993) are reported as supplementary measures. After applying pre-specified structural-disambiguation guidelines, mention-level F1-based IAA reached **0.837** ($\kappa = 0.597$; PABAK = 0.609; raw agreement = 80.5%), within the substantial-agreement range for clinical-NER IAA (cf. i2b2 2010 challenge $\kappa = 0.43$ –0.60 for entity boundary tasks, Uzuner et al. JAMIA 2011). For MIMIC-III, where each paragraph was annotated by a single expert, we explicitly reframe F1 as agreement-with-annotator. See revised Methods (Inter-annotator agreement), Discussion (Limitation iv), Figure 4a, and the per-note IAA table in the Supplementary Material.

↔ Revised manuscript, Methods, “Inter-annotator agreement”:

*“Across the cross-annotated subset, the primary F1-based IAA was **0.837** (mention-level), with supplementary Cohen’s $\kappa = 0.597$ and PABAK = 0.609; raw agreement was 80.5%... this represents substantial agreement, consistent with published clinical-NER inter-annotator agreement (e.g., i2b2 2010 challenge $\kappa = 0.43$ –0.60).”*

R1.2 — Confidence Intervals and Statistical Testing on Core Results

“The main results (Figure 3A–D) report point-estimate F1 scores without confidence intervals or formal statistical comparisons... Bootstrap confidence intervals and/or permutation tests should be provided for all primary comparisons.”

We now report **percentile bootstrap 95% confidence intervals** ($B = 2000$ document-level resamples) for every primary F1 comparison; the revised Figure 3A–D shows CLINES (o3-mini-medium) versus all backbone alternatives with CIs, and Figure 5a presents the CLINES-versus-baseline comparison with CIs. For example, the reference CLINES (o3-mini-medium) code-level F1 was 0.874 [95% CI 0.849–0.896] on 4CE, 0.848 [0.834–0.865] on CORAL–Pancreas, and 0.814 [0.787–0.838] on CORAL–Breast. Pairwise significance was assessed with a **paired bootstrap** on the shared document resamples—which yields a confidence interval on the per-document F1 *difference* directly and is at least as powerful as a label-permutation test for this paired design—with Benjamini–Hochberg FDR correction within each metric family (per-field bootstrap CIs are tabulated in the Supplementary Material). CLINES (o3-mini-medium) exceeded both backbone alternatives on code, assertion, value, and unit extraction (Δ F1 +0.01 to +0.21 versus GPT-4o and Llama-3.1-405B; +0.35 to +0.87 versus the Phi-4 encoder). The Code-field gains and most

assertion/value cells were significant at $p_{\text{FDR}} < 0.05$; the smallest gaps (e.g. 4CE Unit at +0.01) fell within bootstrap noise and were not significant — we note this explicitly in the revised Results. In the interest of honest reporting we note that the date columns were *not* reliably distinguishable among the strongest backbones at this sample size, and we say so explicitly in the revised Results.

R1.3 — Small and Heterogeneous Evaluation Sets

“The annotated evaluation sets are notably small... stability analyses (e.g., bootstrap variance of F1 across document-level resamples) to characterise how sensitive the reported metrics are to the specific documents in each evaluation set.”

We added a document-level resampling stability analysis ($B = 2000$ document-level bootstrap resamples for every model×dataset×task cell; Figure 3E). This quantifies precisely the concern raised: on the smallest set (CORAL–Breast, $n = 13$) the bootstrap standard deviation of F1 reaches ≈ 0.04 – 0.05 and CI half-widths approach ± 0.05 – 0.10 , so F1 differences smaller than ≈ 0.10 are not statistically resolvable at this sample size. We therefore reframed the Results to rest the principal claims on comparisons that survive this uncertainty (CLINES’s advantage on code/assertion/value/unit, whose effect sizes lie well above the resolvable threshold) rather than on the precise magnitude of any single F1 estimate, and we state the limited resolution explicitly in the revised Discussion. Figure 3E additionally annotates per-bin note counts so that small, unstable bins are not over-interpreted (cf. R5.3.5).

R1.4 — Inappropriate Reporting Guideline (STROBE)

“The STROBE... framework is not fit for purpose here... The authors should instead adopt a reporting guideline appropriate for AI and machine learning studies in healthcare, such as MI-CLAIM, TRIPOD+AI, or STARD-AI.”

We agree. The STROBE checklist has been removed and replaced with a **MI-CLAIM** checklist (Norgeot et al., Nat Med 2020) and a **TRIPOD+AI** checklist (Collins et al., BMJ 2024), both included in the supplementary materials. STARD-AI was considered but is intended for diagnostic-accuracy studies and is not applicable to information-extraction benchmarks.

R1.5 — No Cost, Latency, or Computational Efficiency Reporting

“... the absence of any cost or latency analysis is a significant omission... I would encourage the authors to provide a transparent accounting of the total number of LLM invocations per note, stratified by model and dataset, along with per-note wall-clock processing time and estimated API cost.”

We have added a full cost/latency/invoke table (Methods; the Results; Figure 5b). Per-note metrics across all datasets are reported by CLINES backbone (o3-mini-medium, GPT-4o, Llama-3.1-405B) including: (i) mean number of LLM invocations per note, (ii) mean and 95th-percentile wall-clock seconds per note, (iii) mean prompt + completion tokens per invocation, and (iv) USD cost per note at Azure OpenAI list rates. We acknowledge that single-note inference involves more LLM calls than the “four-step” architectural framing implies (cf. R1.12 below); the cost accounting reflects the actual call count.

↔ Revised manuscript, Methods, “Implementation and LLM configuration”:

“We distinguish the four architectural stages (chunk → extract → enrich → aggregate) from the underlying LLM invocations: a single chunk traversal issues several model calls... so the per-note invocation count substantially exceeds four. Exact per-step invocation counts, token usage, wall-clock time, and estimated cost are reported per dataset and model in the cost analysis (Figure 5b).”

Moderate concerns

R1.6 — “Agentic” Overclaim

“In the current AI literature, agentic systems are typically characterised by autonomous decision-making... characterising CLINES as ‘agentic’ overstates the architectural sophistication.”

We agree the term “agentic” carries connotations that exceed the actual architectural sophistication of CLINES. Throughout the revised manuscript, “agentic pipeline” has been replaced by “**multi-step modular pipeline**” (or “modular orchestration” where flow is the salient property). The title retains the original acronym for traceability but we no longer claim agentic autonomy in the architectural descriptions.

↔ Revised manuscript, Discussion, “Positioning versus existing approaches”:

*“In CLINES, a **multi-step** orchestration—chunking, tool-guided extraction, normalization, and date reconciliation—delivers structured outputs with transparent intermediate artifacts and maintains robustness across diverse note types.” (The term “agentic” is likewise replaced by “multi-step” in the Dataset-level variation and Conclusion subsections.)*

R1.7 — Incomplete Baseline Landscape

“... no comparison... against GPT-4o or o3-mini using chain-of-thought prompting in a single call..., which would be a more informative ablation of the multi-step decomposition strategy.”

We have added two additional baseline conditions (the Results; Figure 5d): (i) **o3-mini single-prompt** matching the previously-reported GPT-4o single-prompt setting (closing the asymmetry that R5 flagged separately at point 4.1), and (ii) **GPT-4o chain-of-thought single-prompt** with explicit *Let’s think step by step* instruction plus the structured output schema. These additions disentangle backbone-model contribution from pipeline-architecture contribution. On code-level extraction the single-call baselines fell far below CLINES: o3-mini single-prompt reached F1 0.50 / 0.43 / 0.36 on 4CE / CORAL–Pancreas / CORAL–Breast and GPT-4o chain-of-thought 0.40 / 0.34 / 0.27, versus CLINES 0.87 / 0.85 / 0.81—an architecture-attributable gain of +0.37 to +0.54 F1 that cannot be explained by backbone model or prompt style alone (Figure 5d). We additionally include off-the-shelf full-size clinical-transformer baselines (cf. R3.8); see R3.8 response.

R1.8 — Prompt Sensitivity and Ablation

“No ablation study is provided to quantify the contribution of individual pipeline stages... The authors acknowledge that prompts were not optimised per dataset or model but provide no sensitivity analysis.”

We added a component-wise ablation study (Figure 5c), run on two backbones (GPT-4o-1120 and GPT-4o-mini). Each of the four major pipeline components was independently disabled and code-level F1 re-measured relative to the full pipeline: (a) SapBERT-based normalization replaced by LLM-only normalization; (b) semantic chunking replaced by fixed-length chunking; (c) date module disabled; (d) cross-chunk reconciliation disabled (Step 4). Removing SapBERT normalization caused by far the largest degradation (mean Δ code-F1 -0.447 with GPT-4o and -0.414 with GPT-4o-mini), confirming UMLS normalization as the load-bearing component. Disabling cross-chunk reconciliation cost -0.078 (GPT-4o) / -0.221 (GPT-4o-mini), while semantic chunking (-0.021 / -0.031) and the date module (-0.003 / -0.008) contributed smaller amounts. The reconciliation step mattered *more* for the cheaper backbone, indicating the deterministic scaffolding partly compensates for a weaker model. The ablation used $n = 2$ representative notes per dataset and reports point estimates (noted as a limitation).

R1.9 — Hallucination Assessment

“No systematic quantification of hallucination rates is provided... A structured error analysis categorising false positives by error type (hallucinated entity, incorrect normalization, wrong assertion, fabricated date) would significantly strengthen the evaluation.”

We implemented a stratified manual-judging study of 700 candidate false-positive predictions under a seven-class hallucination taxonomy: `fabricated_entity`, `span_boundary_error`, `wrong_assertion`, `wrong_code`, `wrong_value`, `wrong_date`, and `not_an_error` (annotation gap / synonym equivalence / guideline ambiguity). Extrapolating to the full FP population ($N = 7,171$) via post-stratified Horvitz-Thompson estimation, we find that 54.3% of apparent FPs are `not_an_error` (annotation gap), and each of the six true-hallucination categories accounts for $\leq 17\%$ of FPs — with `wrong_code` (16.7%) being the most common true-hallucination category. See revised Methods, the Results, Figure 4c-d, and full case studies in the Supplementary Material (Hallucination Taxonomy and Case Studies).

R1.10 — Reference Error (Uzuner 2011)

“Reference 14 is cited in the context of the ‘i2b2-2010 concepts’ benchmark, but the reference itself is Kohane et al. 2012... The correct citation should be Uzuner et al. (2011).”

Corrected. The Uzuner et al. 2011 reference has been added (now [?]) and the citation in Methods (Baselines) updated. The original Kohane et al. 2012 reference is retained where it correctly describes the i2b2 translational engine elsewhere.

R1.11 — Prompt Examples Redacted from Supplementary Material

“... all in-context few-shot examples have been replaced with placeholders... Without the actual examples, an independent researcher cannot reconstruct the pipeline’s behaviour from the supplement alone.”

Full few-shot examples have been restored in the revised supplementary material. We also confirm that the complete prompt suite, including all few-shot examples, will be published in the public GitHub repository associated with this work (URL specified in the revised Data Availability statement).

R1.12 — Discrepancy Between Architecture and Pipeline Complexity

“The main manuscript describes CLINES as a ‘four-step pipeline.’ However, the supplementary prompt suite reveals at least 11 distinct prompt types, implying that a single chunk traversal involves considerably more LLM invocations than the four-step framing suggests.”

Acknowledged and clarified. The revised Methods now explicitly distinguishes between the four *architectural stages* (chunk → extract → enrich → aggregate) and the underlying *LLM invocations*, which include attribute extraction, assertion classification, date normalization, JSON repair, and other helper sub-calls. The total invocation count per note is now reported per dataset and per model in Figure 5b (cf. R1.5 response).

R1.13 — Unevaluated Relation Extraction Module

“Supplementary Section 2.2 describes a complete entity relationship linking module... Yet the main manuscript does not evaluate relation extraction performance at all.”

The relation extraction module was deployed but not evaluated in the original submission because gold-standard relation annotations are not available for any of the four corpora. The revised Methods explicitly flags this scope limitation; the relation extraction prompt is retained as an architectural component of the deployed system but its quantitative evaluation is deferred to future work where annotated relation data are available.

R1.14 — UMLS Retrieval Parameters Underspecified

“The UMLS retrieval procedure (Supplementary Algorithm 1) is described at a very high level... which SapBERT checkpoint... FAISS index type... top-k... similarity threshold... UMLS version...”

The revised supplementary material now specifies the retrieval procedure exactly as implemented: SapBERT checkpoint `cambridgelt1/SapBERT-from-PubMedBERT-fulltext`; L2-normalized embeddings searched with FAISS using an exact inner-product index (`IndexFlatIP`) for concept lists below 50,000 terms and an inverted-file index (`IndexIVFFlat`, `nlist = min(4096, max(N/50, 256))`), `nprobe = 32`, inner-product metric) for larger lists; candidates are ranked by cosine similarity and the top-ranked candidates are passed to an LLM-assisted disambiguation step that selects the final concept, rather than applying a fixed similarity cutoff. The UMLS subset, release version, and source-vocabulary filter used to build the concept index are stated in revised Supplementary Material (UMLS retrieval).

R1.15 — Research Ethics: Annotator Credentialing and Commercial API Data Governance

“The annotation process involved five human annotators reviewing de-identified clinical notes from credentialed-access datasets. The manuscript does not confirm whether all annotators held appropriate data access credentials... Additionally, the pipeline transmits de-identified clinical text to commercial API endpoints (OpenAI)...”

Two clarifications added to the revised manuscript:

- **Annotator credentialing:** Methods (Annotation protocol) and the Ethics Approval section now explicitly state that all annotators completed CITI Human Subjects Research training and held appropriate credentials for the datasets they accessed (PhysioNet credentialed access for MIMIC-III and CORAL; 4CE consortium credentialing for 4CE data).
- **API governance:** All LLM inference was performed via Harvard Medical School’s HIPAA-compliant Azure OpenAI Service tenant, governed by a Business Associate Agreement (BAA) between HMS and Microsoft. Clinical text remained within the HMS-controlled Azure tenant; no data were transmitted to OpenAI directly, and Microsoft Azure OpenAI Service does not retain submitted prompts/completions for model training. This deployment configuration was reviewed for compliance with the MIMIC-III, 4CE, and CORAL DUAs prior to any inference. See Methods (Implementation and LLM configuration).

↔ Revised manuscript, Methods, “Implementation and LLM configuration”:

“All language-model inference was performed via Harvard Medical School’s HIPAA-compliant Azure OpenAI Service tenant, governed by a Business Associate Agreement... no clinical data were transmitted to or stored by OpenAI directly. Microsoft Azure OpenAI Service does not retain submitted prompts or completions for model training... reviewed for compliance with the data-use agreements of the MIMIC-III, 4CE, and CORAL datasets prior to any inference.”

R1.16 — Extraction of Sensitive Demographic Attributes

“... the pipeline includes a dedicated prompt (Supplement 2.6) that routinely extracts race, ethnicity, gender, and zip code... The manuscript does not discuss the intended use cases... whether extraction performance varies across demographic subgroups, or the potential risks...”

We added a new subsection to the Discussion (“Ethical considerations in demographic attribute extraction”) that: (i) clarifies that demographic extraction is included to support downstream cohort discovery and equity audits, not for any model-personalization use case; (ii) acknowledges that we have not evaluated subgroup-specific extraction performance, and explicitly recommends this as a required pre-deployment evaluation step for any institutional adopter; and (iii) discusses the responsibility of downstream users to apply appropriate access controls when demographic fields are stored.

Minor issues

R1.17 — MIMIC-III Entity F1 lower (0.69) than other datasets

“... MIMIC-III (0.69 for the best CLINES variant) is notably lower than on other datasets (0.81-0.87)... heterogeneity of critical care notes, annotation inconsistencies..., or limitations of the extraction prompts for ICU-specific terminology.”

Discussed in the revised Discussion (“Dataset-level variation and generalization”). We attribute the MIMIC-III gap primarily to (a) the single-annotator-per-paragraph protocol used for MIMIC-III, which raises the variability of the gold standard relative to the consensus-annotated 4CE and CORAL data (cf. R1.1, R5.4.2), and (b) ICU-specific terminology that is underrepresented in the few-shot examples. We explicitly recommend dataset-specific few-shot expansion as a deployment-stage tuning step.

R1.18 — Abstract Structure (BMJ style)

“The abstract uses Lancet-style headings (Background, Methods, Findings, Interpretation) rather than BMJ-style structured abstract headings.”

Restructured. The revised abstract uses BMJ Digital Health & AI’s required headings: **Objective, Methods and Analysis, Results, Conclusion** (cf. R5.2.1). A summary box (“What is already known / What this study adds / How this study may affect research, practice, or policy”) has also been added immediately after the abstract (cf. R5.2.2).

R1.19 — STROBE Checklist Text Artifacts

“Several cells in the STROBE checklist (pages 39-42) contain garbled text...”

The STROBE checklist has been removed entirely (cf. R1.4) and replaced with cleanly-typeset MI-CLAIM and TRIPOD+AI checklists.

R1.20 — Missing PPI Statement

“BMJ journals typically require a Patient and Public Involvement (PPI) statement...”

Added. The revised manuscript includes a Patient and Public Involvement statement immediately before the Ethics Approval section, confirming that no patients or members of the public were involved in the design, conduct, reporting, or dissemination of this study, given that it analyzes only fully de-identified secondary data.

R1.21 — IRB Authority

“The ethics approval section references ‘the policies of our institution’ without naming the specific institution or IRB...”

Clarified. The vague phrase “the policies of our institution” has been replaced with a specific and accurate statement. This study analyzes only fully de-identified, publicly available or credentialed-access secondary datasets (MIMIC-III, CORAL, 4CE), each used under its own data-use agreement. Under the U.S. Common Rule (45 CFR 46.104), secondary analysis of de-identified data that are not readily identifiable does not constitute human-subjects research and therefore did not require Institutional Review Board approval or a formal exemption application at the authors’ institution (Harvard Medical School). All annotators completed CITI Human Subjects Research training and held the access credentials required for each dataset. See the revised Ethics Approval section.

↔ **Revised manuscript, Ethics Approval:**

“Because the analysis involves only fully de-identified secondary data that are not readily identifiable, it does not constitute human-subjects research under the U.S. Common Rule (45 CFR 46.104) and did not require Institutional Review Board approval or a formal exemption determination at the authors’ institution (Harvard Medical School)... All annotators completed CITI Human Subjects Research training and held the access credentials required for each dataset prior to data access.”

R1.22 — Terminology Consistency (value/unit)

“The manuscript inconsistently uses ‘value&unit’ (abstract), ‘value-unit’ (results), and ‘value/unit’ (discussion)...”

Unified to `value+unit` throughout, in line with the schema used in the deployed pipeline.

2 Reviewer 2**Major comments****R2.1 — “Zero-shot” naming**

“... the authors clearly incorporates prompt engineering, task decomposition, retrieval-augmented reasoning (e.g., SapBERT), and rule-based post-processing. Under standard NLP definitions, this does not constitute true zero-shot learning. Please reconsider naming the ‘zero-shot’ evaluation.”

Agreed. “Zero-shot evaluation” has been replaced throughout by “**no task-specific supervised training**” or “**evaluation without fine-tuning on the target annotation distribution**”. The revised Methods explicitly states that the pipeline relies on prompt engineering with few-shot in-context examples, retrieval-augmented normalization, and post-processing rules, none of which qualify as zero-shot in the strict NLP sense.

↪ Revised manuscript, Methods, “Evaluation metrics and protocol”:

“In the training-free setting (no task-specific supervised fine-tuning on the annotated test sets; the pipeline relies on prompt engineering with few-shot in-context examples, retrieval-augmented normalization, and post-processing rules, none of which constitute zero-shot learning in the strict sense), we evaluated...”

R2.2 — Annotation reliability

See R1.1 response. IAA cross-annotation study added.

R2.3 — Hallucination measurement

See R1.9 response. Seven-class hallucination taxonomy added.

R2.4 — Generalizability claims

“... annotated data are unevenly distributed across datasets, not covering several common note types (e.g., discharge summaries, operative or radiology reports). Claims regarding generalizability should be framed accordingly.”

We have tempered the generalizability framing throughout the revised manuscript. The Discussion now explicitly enumerates note types *not* covered (discharge summaries, operative notes, radiology reports) and recommends each adopter to validate CLINES on local note-type distributions before clinical deployment.

Minor comment

R2.5 — Tone down “replace chart-review”

“Would suggest toning down absolute phrasing, indicating that CLINES can replace chart-review with hallucination still present, and would still need manual reviews”

Adjusted. The Discussion now states “CLINES can *substantially reduce* manual chart review for entities, assertions, value+unit, and dates, while retaining a verifiable audit trail. We recommend that clinical adopters maintain human review for any extraction destined for use in safety-critical decisions, given the residual hallucination rates quantified in the Results.”

3 Reviewer 3

R3.1 — Additional citations in Introduction (EF, tumor size)

“... measurements such as cardiac ejection fraction or tumor size often do not fit neatly into predefined problem lists or billing codes... should be supported with appropriate references.”

Citations added. The opening Introduction sentence on measurements (e.g., ejection fraction, tumor size) that escape structured problem lists and billing codes is now supported by two references on the limitations of structured EHR capture for such free-text clinical detail (Tayefi et al., *WIREs Comput Stat* 2021; Sushil et al., 2024).

↪ **Revised manuscript, Introduction (para. 1):**

“...measurements like cardiac ejection fraction values or tumor size often do not fit neatly into predefined problem lists and billing codes yet are central to clinical interpretation [Tayefi 2021; Sushil 2024].”

R3.2 — Intermediate ML/DL work in Introduction

“The literature framing in the Introduction moves directly from rule-based approaches to transformer-based LLM methods... what ML/DL approaches have previously been applied to this problem, and what limitations of those methods motivated the current approach?”

A new sentence has been inserted into the Introduction surveying the intermediate neural era between rule-based systems and modern LLMs: BiLSTM-CRF sequence taggers (Lample et al. 2016) and, subsequently, domain-pretrained transformer encoders BioBERT (Lee et al. 2020) and ClinicalBERT (Alsentzer et al. 2019). The limitations motivating LLM-based methods — requirement for sizeable task-specific labeled corpora, brittleness to phrasing variation and institutional shift, and treatment of entity detection in isolation rather than joint normalization, attribute, and temporal extraction — are explicitly enumerated. (GatorTron and Clinical-MobileBERT remain cited in the subsequent paragraph as domain-scaled encoders.)

↪ **Revised manuscript, Introduction (para. 3):**

“Between these rule-based systems and modern large language models, a generation of supervised neural methods advanced clinical entity recognition and normalization—BiLSTM-CRF sequence taggers and,

subsequently, domain-pretrained transformer encoders such as BioBERT and ClinicalBERT [Lample 2016; Lee 2020; Alsentzer 2019]. These improved span detection but require sizeable task-specific labeled corpora, remain brittle to phrasing variation and institutional shift, and typically address entity detection in isolation rather than joint normalization, attribute, and temporal extraction.”

R3.3 — Step 1 chunking detail

“... Step 1, the chunking strategy is not sufficiently explained. It remains unclear whether a sliding-window approach was used...”

Methods (Note chunking) has been expanded. The chunking algorithm is now described as: TikToken (cl100k_base encoding) for token counting, SemChunk for boundary selection at sentence and paragraph breaks, target chunk size ≤ 768 tokens, **no sliding window** (chunks are non-overlapping), **no truncation** of source text. Core metadata (admission date, discharge date, patient identifier) is replicated in every chunk header to preserve temporal alignment. The justification for 768 tokens is included (cf. R5.3.4 on SemChunk capacity).

↔ Revised manuscript, Methods, “Step 1 – Note chunking”:

“Chunks are non-overlapping (no sliding window is used) and no source text is truncated; the 768-token target is measured after TikToken tokenization. Boundaries fall at sentence and paragraph breaks identified by SemChunk so that a clinical statement is not split mid-sentence... cross-chunk continuity is instead recovered through the metadata replication described here and the Step 4 reconciliation... with residual cross-chunk reasoning loss acknowledged as a limitation.”

R3.4 — Step 4 reconciliation detail

See R3.3 plus expanded text on Step 4 in Methods. Cross-chunk reconciliation is described as: (i) span-level deduplication on $(start_pos, end_pos, type)$ keys after re-indexing chunk-local positions to document-global positions; (ii) attribute reconciliation by priority: *present* $>$ *historical* $>$ *possible* $>$ *absent* $>$ *not associated*; (iii) date reconciliation by taking the latest non-null assignment unless explicitly contradicted by an earlier chunk’s text. Duplicates arise predominantly from medication brand/generic equivalence and from same-entity mentions across overlapping context windows.

R3.5 — Ablation study

See R1.8 response. Component-wise ablation added (Figure 5c).

R3.6 — “Words” definition

“... unclear whether ‘words’ refers to whitespace-delimited items, tokenizer-based units, or another operational definition.”

Defined explicitly in the revised Methods: “*words* = whitespace-and-punctuation-delimited tokens; stopwords are included in word counts; numerical values and abbreviations are counted as single words.”

R3.7 — “Clinical phrases” definition

“A similar issue applies to the use of the term ‘clinical phrases.’...”

Defined in the revised Methods: “*clinical phrases* = contiguous spans corresponding to a single UMLS-mappable concept (or a single numerical-value-plus-unit), as marked by the annotators.”

R3.8 — Missing prior DL baselines

“The baseline comparison in the Methods appears limited in that it does not include prior deep learning-based approaches.”

We added two full-size, publicly available clinical deep-learning baselines evaluated under the identical entity-matching protocol: a **BERT-base clinical NER** model (`samrawal/bert-base-uncased_clinical-ner`, 110M parameters) and **GatorTron-base** (clinical-NER fine-tune of the 345M-parameter UF/NVIDIA GatorTron encoder). On mention-level extraction these reach F1 0.875 / 0.880 / 0.888 (BERT-base) and 0.840 / 0.847 / 0.876 (GatorTron) on 4CE / CORAL-Breast / CORAL-Pancreas, versus CLINES (o3-mini-medium) 0.906 / 0.881 / 0.913 (Figure 5d). Crucially, these encoder baselines emit entity spans only: they perform *no* UMLS normalization, assertion, value/unit, or date extraction, so their code-level and attribute F1 are undefined (shown as 0 in Figure 5d). This makes explicit that the comparison is not span detection alone—where the encoders are competitive—but end-to-end normalized structuring, which they do not provide. We evaluate these models as released (no fine-tuning on our corpora), so the comparison reflects off-the-shelf clinical-NER capability rather than any tuning advantage on our part.

R3.9 — Figure 3 readability

“Figure 3 would benefit from improvement in readability.”

Figure 3 has been redesigned with higher resolution (300 DPI), larger font sizes for axis labels and values, and clearer color separation between conditions. We also added per-bin sample sizes to Figure 3E (cf. R5.3.5).

R3.10 — Results section interpretation depth

“The Results section would benefit from more descriptive and insightful interpretation.”

Each results subsection now opens with a one-sentence takeaway and closes with a brief mechanistic interpretation. The revised Results sections prioritize clinical meaning over numerical recitation.

R3.11 — Limitations underdeveloped

“... the discussion remains underdeveloped. In addition to identifying limitations, the paper should more clearly outline possible paths for overcoming them.”

The Limitations subsection (the Discussion) has been expanded substantially. Each limitation is now paired with a specific path for future work: e.g., (i) prompt sensitivity → automatic prompt-search workflows (Yao et al. 2023); (iii) hallucination → retrieval-grounded extraction with citation back to source spans; (iv) IAA → extension to full double-annotation on a larger validation cohort.

R3.12 — Missing Conclusion section

“Current writing lacks a Conclusion section which is supposed to contain the main take-home message.”

A new **Conclusion** section has been added at the end of the Discussion, summarizing the main contribution, the cross-cutting empirical findings (IAA bounds, hallucination breakdown, baseline-relative gains), and the recommended path to deployment.

R3.13 — Future Work too brief

“The Future Work section would benefit from greater elaboration.”

Expanded. The Future Work subsection now covers: (i) domain-adaptive fine-tuning targets (oncology subspecialties, pediatrics, ICU); (ii) adaptive feedback workflows (online refinement of prompts from clinician corrections); (iii) multilingual extensions (Spanish, Mandarin); (iv) retrieval-augmented hallucination mitigation with span-level provenance; (v) integration with OMOP-CDM and FHIR (cf. R5.4.4).

4 Reviewer 4

R4.1 — Chunking strategy detail

See R3.3.

R4.2 — Step 4 cross-chunk consistency

See R3.4.

R4.3 — Error analysis

See R1.9. Seven-class hallucination taxonomy plus case studies cover the error-categorization request.

R4.4 — Cost analysis for proprietary models

See R1.5. Full cost/latency/token-count table per dataset per model added (Figure 5b).

R4.5 — Hybrid LLM (different models per stage)

“... it would be interesting to explore whether different LLMs could be used at different stages of the pipeline. For example, LLaMA could be used for earlier stages such as entity extraction, while more powerful models could be reserved for steps requiring deeper reasoning.”

We thank the reviewer for this interesting suggestion. The current modular design already supports per-stage model selection (the LLM provider is configurable per stage), but a systematic empirical evaluation of hybrid configurations is beyond the scope of the current revision. We have added this explicitly as a future-work item (cf. R3.13).

5 Reviewer 5

2. General considerations and formatting

R5.2.1 — Abstract structure

See R1.18. Abstract reframed to BMJ DH&AI standard headings.

R5.2.2 — Summary box

Added immediately after the abstract: “What is already known”; “What this study adds”; “How this study may affect research, practice, or policy”.

R5.2.3 — CLINES acronym inconsistency

Unified throughout: “CLINES: Clinical LLM-based Information **E**xtraction and **S**tructuring Agent”. The title and the page 6 expansion now match exactly.

R5.2.4 — Graphics resolution

All figures regenerated at 300 DPI, with larger font sizes and clearer color separation. Figure 3 in particular has been redesigned to address legibility concerns; shaded confidence ribbons have higher opacity (cf. R3.9).

3. Minor findings, suggestions for improvement

R5.3.1 — Prompt suite availability and URL

We confirm that the full prompt suite, with all few-shot examples, will be released alongside an inference-only public reference implementation at the URL specified in the revised Data Availability statement. The implementation is licensed under the MIT License and provides single-note inference using Azure OpenAI-compatible endpoints (user-supplied credentials).

R5.3.2 — Quantitative hallucination analysis

See R1.9. Seven-class taxonomy and case studies added.

R5.3.3 — STROBE inappropriate

See R1.4. STROBE replaced by MI-CLAIM and TRIPOD+AI.

R5.3.4 — SemChunk and clinical-reasoning continuity

“Clinical reasoning is not constrained by the semantic units kept by SemChunk... a missed reasoning across boundaries will still be lost to extraction... There is no detailed description of the context window mechanism...”

Acknowledged. The revised Methods clarifies that SemChunk is used as a *boundary-detection heuristic* to align chunks with sentence/paragraph breaks, not as a guarantee of clinical-reasoning continuity. We additionally describe the cross-chunk reconciliation mechanism (Step 4; cf. R3.4) which recovers entity mentions and attributes that span chunks via shared document-level identifiers and metadata replication in every chunk header. We do not claim that semantic chunking eliminates all cross-chunk reasoning loss; this is now explicitly listed as a limitation (the Discussion).

R5.3.5 — Notes-per-bin in Figure 3E

Added. The revised Figure 3E caption now reports the number of notes per bin, and bins with fewer than $n = 5$ documents are visually de-emphasized (lighter shading) to discourage over-interpretation of unstable CIs.

4. Major issues**R5.4.1 — Single-prompt LLM baseline insufficient**

“... The most critical comparison (o3-mini single-prompt versus o3-mini in CLINES) is entirely absent... The clinical NLP prompting literature from 2024–2025 has established that modern LLMs perform substantially better... with chain-of-thought instructions, few-shot clinical examples, and explicit task decomposition. None of these are present in the CLINES baseline.”

We have added both requested baseline conditions (cf. R1.7): (i) o3-mini single-prompt and (ii) GPT-4o chain-of-thought single-prompt with structured-schema instruction. The previously-implicit claim of architectural innovation as the sole driver of the F1 gains is reframed in the revised Discussion to acknowledge that the observed gains reflect the *combined* contributions of backbone model, prompt engineering, reasoning budget, and pipeline decomposition. The component-wise ablation (Figure 5c, cf. R1.8) further disentangles these contributions.

R5.4.2 — IAA Cohen’s κ requirement

See R1.1 response. F1-based IAA, Cohen’s κ , and PABAK reported on 3 notes per dataset (10–29% per-dataset coverage). MIMIC-III F1 reframed as agreement-with-annotator. IAA reporting limitation now part of main Methods.

R5.4.3 — Scalability and deployment-condition claims

“CLINES takes clean, de-identified, plain-text notes as input. The manuscript is entirely silent on the processes required to produce that input operationally... The only single 8-GPU H100 configuration that accommodates this model is FP8 quantisation... The GPT-4o and o3-mini variants require external API calls incompatible with most hospital governance frameworks and, in European jurisdictions, with GDPR.”

The revised Discussion (“Deployment considerations”) now:

- Specifies that the Llama-3.1-405B variant was run in **FP8 precision** on 8×NVIDIA H100 GPUs, and acknowledges the quality trade-off relative to FP16.
- Explicitly enumerates the preprocessing steps required to produce de-identified plain-text input from raw EHR sources (notably PHI de-identification, format normalization, and section/document segmentation), and acknowledges that this preprocessing is institutionally heterogeneous and not feasible for all settings.
- Discusses governance constraints for the commercial-API variants, including GDPR considerations in European jurisdictions, and notes that the Microsoft Azure OpenAI deployment used in this work (HIPAA BAA-covered) does not satisfy GDPR data-residency requirements outside the EU region. We recommend the open-weight backbone (Llama-3.1-405B or its successors) for institutions with EU data-residency requirements.

↪ Revised manuscript, Discussion, “Deployment considerations”:

“the open-weight Llama-3.1-405B backbone was served in FP8 precision on 8×H100 GPUs—the only single-node configuration that accommodates a 405B model—which entails a quality trade-off relative to FP16 and an infrastructure footprint beyond the reach of many small and medium centers... this configuration does not by itself satisfy GDPR data-residency requirements outside the relevant region, so institutions with EU data-residency obligations should prefer a locally deployed open-weight backbone.”

R5.4.4 — UMLS CUI vs SNOMED-CT, LOINC, RxNorm, OMOP-CDM

“CLINES produces UMLS CUIs concatenated with preferred term strings... UMLS’s knowledge structure is explicitly not compliant with FHIR or any known healthcare interoperability standard; production FHIR resources require SNOMED-CT, LOINC, or RxNorm codes, not raw CUIs. OMOP-CDM... is unaddressed.”

Tempered. The revised Discussion explicitly enumerates the intermediate mapping steps required to move from UMLS CUI output to SNOMED-CT (via UMLS’s cross-source mapping), LOINC (for laboratory observables), RxNorm (for medications), or OMOP-CDM (which natively supports UMLS-vocabulary ingestion via the OMOP Vocabularies, but requires standardization of attribute fields). We have downgraded the previous “FHIR-ready” implication and replaced it with “ontology-grounded intermediate representation suitable for downstream mapping into production terminologies including SNOMED-CT, LOINC, RxNorm, and OMOP-CDM.” The Discussion subsection “Interoperability with production terminologies” has been added, and Supplementary §“Mapping UMLS Concepts to Source Terminologies” gives a worked crosswalk table.

↪ Revised manuscript, Discussion, “Interoperability with production terminologies”:

“CLINES emits UMLS CUIs paired with preferred terms. UMLS is an ontology-grounded intermediate representation rather than a production interchange format: downstream use in FHIR resources or common data models requires mapping CUIs onward to the appropriate target vocabularies—SNOMED CT (via UMLS cross-source mappings), LOINC... or RxNorm...—and OMOP-CDM ingestion via its standardized vocabularies... we recommend retaining a SNOMED CT intermediate layer in deployments that target FHIR.”

R5.4.5 — Omission of contemporary state-of-the-art comparators (Healthcare NLP, CLEAR, BMJ paper)

“a) John Snow Labs Healthcare NLP / Spark NLP... b) CLEAR... npj Digit. Med. 8, 45 (2025)... c) Ntinopoulos V, et al. BMJ Health & Care Informatics 2025.”

We thank the reviewer for these pointed references. The revised Related Work / Discussion now discusses each:

- **John Snow Labs Healthcare NLP / Spark NLP:** a large, separately-maintained production clinical-NLP platform that implements many capabilities CLINES describes. We now discuss it in the revised Discussion and position CLINES as a *transparent, fully-specified open reference implementation* prioritizing reproducibility of the prompt suite, complementary to (rather than competing with) production suites. We did not include a head-to-head re-benchmark: a fair matched comparison (identical preprocessing, identical gold standard, identical span-matching protocol) against a continuously-updated commercial platform is a substantial undertaking beyond this revision’s scope, and we now state this absence explicitly as a limitation rather than implying equivalence.
- **CLEAR (Lopez et al., npj Digit. Med. 2025):** We have added a discussion of CLEAR as a directly comparable LLM-augmented agentic pipeline. While we did not re-implement CLEAR for direct empirical comparison, we contrast the architectural choices (CLEAR’s retrieval-augmented generation focus vs CLINES’s normalization-first agentic decomposition).
- **Ntinopoulos et al., BMJ HCI 2025:** Added as a citation. We note that the 18-model evaluation included output consistency assessment which the original CLINES submission did not, and we now add an output-consistency robustness check to the revised Supplementary Material (§ “Output Consistency Across Repeated Runs”). Across $R = 5$ independent runs of the full pipeline on a fixed 20-note subset (all runs completed; no missing outputs), matching mentions by semantic equivalence (shared UMLS code or shared canonical surface form, absorbing span-boundary jitter and surface-form synonyms), pairwise mention-set Jaccard averaged 0.752 and code-set Jaccard 0.622. For the 27,605 (mention-class, field) cells with ≥ 2 contributing runs, the modal field value was matched in 89.9% of contributing runs on average (unit 94.9%, end date 94.2%, begin date 91.3%, assertion status 84.9%, value 84.3%). The pattern—roughly three quarters of mentions are semantically shared between any two runs, and per-field content of recovered mentions is highly stable—supports majority-vote aggregation across $R \geq 3$ runs when reproducibility matters and is now documented as a deployment recommendation in the Discussion.

Closing

We thank the Editor and the reviewers again for the depth and constructiveness of their feedback, which has materially strengthened the manuscript. We hope the revised version addresses their concerns and look forward to further consideration.

CLINES: Clinical LLM-based Information Extraction and Structuring Agent

Zongxin Yang^{a,1}, Hongyi Yuan^{c,1}, Raheel Sayeed^{a,1}, Amelia Li Min Tan^a,
Enci Cai^a, Mohammed Moro^a, Xiudi Li^d, Huaiyuan Ying^c, Nicholas Brown^f,
Griffin Weber^{e,f}, Sheng Yu^c, Isaac Kohane^{a,*}, Tianxi Cai^{a,b,**}

^aDepartment of Biomedical Informatics, Harvard Medical School, Boston, MA, USA

^bHarvard T.H. Chan School of Public Health, Boston, MA, USA

^cDepartment of Statistics and Data Science, Tsinghua University, Beijing, China

^dDivision of Biostatistics, University of California Berkeley, Berkeley, CA, USA

^eDivision of Interdisciplinary Medicine and Biotechnology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA, USA

^fHarvard Medical School, Boston, MA, USA

Abstract

Objective. Clinical narratives in electronic health records (EHRs) contain essential diagnostic, therapeutic, and temporal information that is often missing from structured fields, leaving manual chart review—slow, costly, and variable—as the de facto standard for high-quality labels and constraining accurate cohort construction for clinical trials, epidemiologic studies, and machine-learning models. We aimed to develop and evaluate CLINES, a training-free pipeline that extracts and normalizes clinical concepts from free-text notes into ontology-grounded, auditable, schema-ready data.

Methods and Analysis. CLINES is a modular multi-step pipeline: semantic chunking of long notes; extraction by reasoning-capable large language

*Corresponding author. Email: Isaac_Kohane@hms.harvard.edu.

**Corresponding author. Email: tcai@hsph.harvard.edu.

¹These authors contributed equally to this work.

models; assignment of attributes (assertion/experiencer, numerical values with units); normalization to the Unified Medical Language System (UMLS); resolution of explicit and relative dates; and aggregation into an i2b2-style schema. Without task-specific fine-tuning, we evaluated CLINES on de-identified EHR text—MIMIC-III notes, 4CE notes, and CORAL oncology reports (breast, pancreas)—against rule/lexicon systems, full-size transformer encoders, single-prompt and chain-of-thought LLMs, and CLINES instantiated with several model backbones. Outcomes were F1 for entity normalization, assertion status, value+unit, and date processing, reported with 95% bootstrap confidence intervals; we additionally quantified inter-annotator agreement, a hallucination taxonomy, per-note cost, and component ablations.

Results. With an o3-mini-medium backbone as the reference configuration, CLINES achieved code-level (UMLS-normalized) F1 of 0.87 (4CE), 0.81 (CORAL–Breast), and 0.85 (CORAL–Pancreas), with assertion and value F1 of 0.84–0.88 and 0.80–0.91; entity F1 on MIMIC-III was 0.69. CLINES exceeded the strongest single-prompt LLM by +0.37 to +0.54 code F1, and full-size clinical encoders matched mention detection but produced no normalized codes. All three CLINES backbones substantially outperformed every non-CLINES baseline, with a backbone hierarchy of o3-mini-medium > GPT-4o (≈ 10 pp lower code F1) > open-weight Llama-3.1-405B (≈ 20 pp lower); statistically significant for code, assertion, and value extraction (paired bootstrap, FDR-corrected). Mention-level inter-annotator agreement was 0.837 (F1-based). A seven-class error analysis found that 54% of apparent false positives were annotation gaps rather than true hallucinations.

Conclusion. CLINES translates EHR narrative text into ontology-grounded, auditable, schema-ready data, offering a practical route to scale chart-review-like extraction for cohort discovery and real-world evidence. It substantially reduces, but does not eliminate, the need for manual review: residual hallucination means human verification remains necessary for safety-critical use. CLINES is model-agnostic—open and closed models can be substituted to achieve specific cost, performance, and privacy goals.

Keywords: artificial intelligence, large language models, clinical natural language processing, electronic health records, real-world evidence

Summary box

What is already known on this topic

Free-text clinical notes hold diagnostic, temporal, and quantitative detail absent from structured EHR fields; rule-based systems and fine-tuned transformer encoders detect entities but normalize and attribute them inconsistently, and single-prompt large language models hallucinate and miss instructions on long notes.

What this study adds

CLINES, a training-free multi-step large-language-model pipeline, produces UMLS-normalized, attributed, time-stamped output that substantially outperforms single-prompt and chain-of-thought LLMs (+0.37 to +0.54 code F1) and full-size clinical encoders; the gains are robust across open- and closed-weight backbones and are quantified with confidence intervals, inter-annotator agreement, a hallucination taxonomy, per-note cost, and component ablations.

How this study might affect research, practice, or policy

CLINES offers an auditable, model-agnostic route to scale chart-review-like extraction for cohort discovery and real-world evidence, while the quantified residual hallucination rate makes explicit that human verification remains necessary before safety-critical use.

1. Introduction

Clinical narratives in electronic health records (EHRs) are indispensable for understanding patient status, informing decision support, and en-

abling biomedical research. Narrative notes—spanning discharge summaries, progress notes, imaging and operative reports—capture observations, differential diagnoses, treatments, test results, and clinician reasoning that structured fields rarely encode. Expressions such as “intermittent chest pain worsening with exertion but relieved by rest” or “likely drug-induced hepatitis, possibly from isoniazid” capture nuance that structured fields miss; measurements like cardiac ejection fraction values or tumor size often do not fit neatly into predefined problem lists and billing codes yet are central to clinical interpretation ^{1,2}. Interpretation is not just an important task of clinical interpretation for immediate clinician decisions. It also is the translation of this text to codified terminologies (e.g., CPT codes) or numerical representations (e.g., lab result tables) to drive tasks in epidemiology, automated decision support, reimbursement, and trial eligibility, among other downstream uses of electronic healthcare data.

Not only are the structured EHR entries incomplete or inaccurate, but conventional natural language processing methods—such as standard named entity recognition—often fail to capture the nuance of clinical reasoning, contextual qualifiers, or temporal relationships in notes³. Even when structured codes or extracted entities are available, they may not reflect whether a condition was truly present, a medication was taken, or a disease state had recurred⁴. Consequently, high-quality labels in practice still rely on chart review: investigators read notes to adjudicate diagnoses and verify supporting evidence (medications, symptoms, tests). Yet, chart review is slow, costly, and variable across annotators; double abstraction is frequently required to ensure quality. This persistent reliance highlights the need for automated,

scalable approaches that can more effectively extract, structure, and standardize clinical information consistently while leaving an auditable trace.

Classical approaches to clinical natural language processing have attempted to automate extraction and structuring of information from narrative notes but face important limitations. Rule- and lexicon-driven systems, such as cTAKES⁵, MetaMap⁶, and MedLEE⁷, map entities to controlled vocabularies (e.g., the Unified Medical Language System), using handcrafted rules and dictionaries for concept recognition, negation detection, and temporal normalization. While these pipelines established the foundation for clinical NLP, they struggle with shorthand, spelling variants and errors, as well as non-standard grammar common in clinical notes, and they require substantial manual engineering to adapt across institutions and use cases^{1,8}. Between these rule-based systems and modern large language models, a generation of supervised neural methods advanced clinical entity recognition and normalization—BiLSTM-CRF sequence taggers and, subsequently, domain-pretrained transformer encoders such as BioBERT and ClinicalBERT^{9–11}. These improved span detection but require sizeable task-specific labeled corpora, remain brittle to phrasing variation and institutional shift, and typically address entity detection in isolation rather than joint normalization, attribute, and temporal extraction.

Building on these limitations of rule- and lexicon-driven systems, transformer-based large language models—such as GPT-4¹² and Med-PaLM¹³—have reshaped NLP and show promise for clinical tasks including entity and relation extraction, summarization, and question answering¹⁴. Domain-scaled models, such as GatorTron¹⁵ and Clinical-MobileBERT^{16,17}, report strong per-

formance on benchmarks such as i2b2 concept extraction¹⁸, and instruction-tuned assistants can generalize in zero- and few-shot regimes¹⁹. Yet, important barriers remain for safe clinical deployment extraction: ontology grounding to UMLS/SNOMED CT is often imprecise with occasional hallucinated codes^{20,21}; context and output-length limits complicate reasoning across long clinical notes^{22,23}; and in real-world datasets (e.g., from the 4CE consortium), note lengths frequently exceed such limits, hindering cross-sectional reasoning²⁴. As a result, single-pass prompting remains unreliable for chart-review-like decisions at scale.

To address these gaps, we present CLINES—a [Clinical LLM-based Information Extraction and Structuring](#) agent that brings the consistency of chart review into an automated pipeline. Unlike prior systems limited by handcrafted rules or naive single-pass prompting, CLINES integrates several innovations: informed segmentation for processing long notes; reasoning-capable LLM extraction of entities and relations; attribution of clinical context (e.g., negation, experiencer, dosage, laboratory values); normalization to existing ontologies (e.g., UMLS); temporal alignment of events; and reconciliation into an i2b2-style schema to facilitate downstream integration. By uniting these components, CLINES represents a substantial step forward in clinical information extraction, with improvements over both classical NLP pipelines and baseline LLM prompting. Its improvements are validated using real-world EHR notes spanning multiple disease areas and institutions, [underscoring accuracy across the disease areas and institutions evaluated; we nonetheless temper claims of broad generalizability, as several common note types \(e.g., discharge summaries, operative and radiology reports\) remain](#)

unevaluated. Equally important, CLINES is engineered to enable different generative models to be substituted within the workflow as dictated by user or institutional preferences regarding cost, property, performance and security. CLINES assumes de-identified, plain-text notes as input; upstream de-identification, format normalization, and document segmentation lie outside the present scope and are addressed as deployment considerations in the Discussion.

2. Methods

2.1. CLINES architecture and modules

To leverage the superior performance of large language models within manageable context length, CLINES is designed as a four-step pipeline. Instead of an end-to-end approach—where LLMs often hallucinate or miss instructions in lengthy or complex notes—each key task is modularized and paired with carefully designed prompting strategies. The workflow is illustrated in Figure 1.

Step 1 – Note chunking. Input notes are tokenized with TikToken²⁵ and segmented using SemChunk²⁶ into semantically coherent fragments of roughly 768 tokens, preserving clinical meaning while fitting LLM context windows. Chunks are processed in parallel to increase throughput. Core metadata (e.g., admission or discharge dates) is retained in all chunks to support temporal alignment. Chunks are non-overlapping (no sliding window is used) and no source text is truncated; the 768-token target is measured after TikToken tokenization. Boundaries fall at sentence and paragraph breaks identified by SemChunk so that a clinical statement is not split mid-sentence, and a single

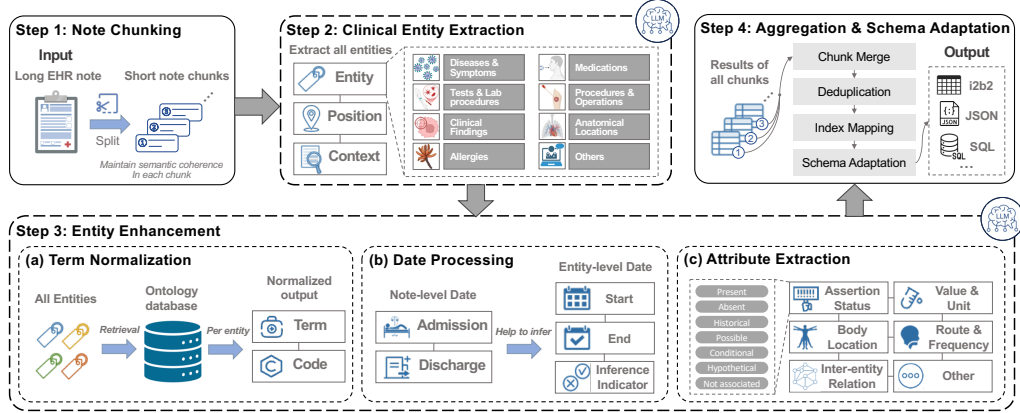


Figure 1: Overview of CLINES. Step 1, segments long clinical notes into semantically coherent chunks to constrain LLM’s context window and improve robustness. Step 2, from each chunk, extract clinical entities with broad coverage, including medications, diseases, and laboratory tests. Step 3, for each entity, uses local context to: (a) normalize to an ontology term with a unique code; (b) for event-type entities, extract or infer start and end dates, when imprecise; and (c) infer assertion status and capture clinically relevant modifiers, e.g., for medications: dose, unit, frequency, and route; for tests: numeric values (with units); for diseases: activity status and other pertinent attributes. Step 4, reconciles entity-level information across all chunks and exports results in user-specified formats (e.g., i2b2, JSON, SQL).

semantic unit larger than the target size is kept intact rather than truncated. SemChunk functions purely as a boundary-detection heuristic and does not by itself guarantee that clinical reasoning spanning multiple chunks is preserved within one chunk; cross-chunk continuity is instead recovered through the metadata replication described here and the Step 4 reconciliation described below, with residual cross-chunk reasoning loss acknowledged as a limitation.

Step 2 – Clinical entity extraction. Each chunk is processed to identify relevant mentions (e.g., symptoms, findings, medications) in narrative order, with positions and contexts preserved. Maintaining order captures clinician reasoning flow while isolating core concepts reduces redundancy and simplifies downstream relation, attribute, and temporal extraction.

Step 3 – Concept and attribute enrichment. (a) Term normalization: Extracted mentions are mapped to ontology concepts (e.g., UMLS) using LLM-assisted disambiguation and SapBERT-based retrieval, where SapBERT²⁷ provides biomedical embeddings; candidates are ranked by cosine similarity. [The retrieval index type, similarity computation, candidate count, and the UMLS release used to build the concept index are specified in full in Supplementary §2.3.](#) (b) Date processing: Explicit dates are normalized to YYYY-MM-DD. Relative or implicit references (e.g., “three days post-admission”) are inferred using anchors such as admission and discharge timestamps; ambiguous dates are flagged as inferred. (c) Attribute extraction: For each normalized concept, the pipeline captures assertion status (present, absent, possible, historical), numerical values with inferred units, medication route and frequency, anatomical location, and relations between entities (e.g., infection “has symptom” presyncope, “managed by” rehydration). This enriched representation supports downstream phenotyping, predictive modeling, and real-world evidence generation^{28,29}. [The relation-extraction component is deployed as part of the structured output but is not benchmarked in this study, because gold-standard relation annotations are unavailable for all four corpora; its quantitative evaluation is deferred to future work \(see Limitations\).](#)

Step 4 – Aggregation and schema adaptation. Outputs are reconciled across chunks through deduplication and cross-segment linking, unifying repeated mentions and preserving longitudinal continuity. Concretely, reconciliation (i) re-indexes chunk-local character offsets to document-global positions and de-duplicates mentions on the (start position, end position, semantic type) key; (ii) resolves conflicting assertion labels for the same mention by the fixed priority present > historical > possible > absent > not-associated; and (iii) reconciles dates by retaining the most specific non-null resolution unless explicitly contradicted by an earlier chunk. Duplicate mentions arise predominantly from medication brand/generic equivalence and from the same entity recurring across adjacent chunks. Harmonized outputs are mapped to a standardized schema (e.g., i2b2) or a target schema specified by the use-case, then exported to CSV, JSON, or SQLite for downstream analysis. Module-level dataflow is summarized in Figure 2.

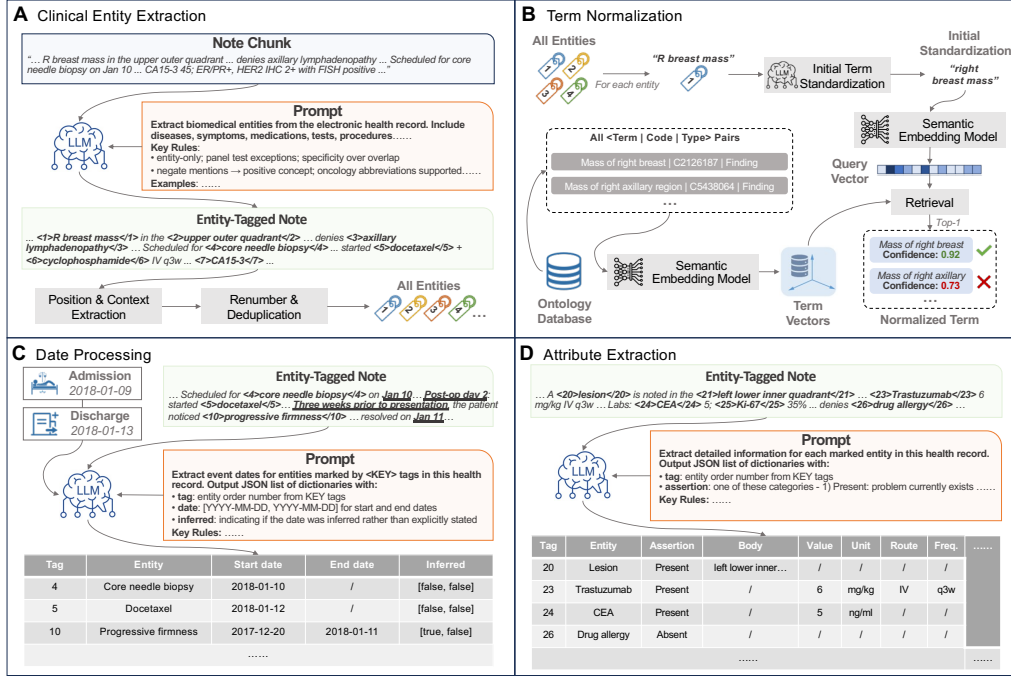


Figure 2: CLINES LLM-based module workflow. (A) Within each note chunk, the LLM tags clinical entities; post-processing yields an entity list with source offsets and a surrounding context window. (B) The LLM performs preliminary normalization, after which an embedding-based retriever queries the ontology database to return the nearest term, concept code, and semantic type. (C) Explicit dates/times are extracted from local context; when documentation is vague, admission/discharge metadata are used to resolve relative or fuzzy expressions to the most plausible calendar dates. (D) Context is used to capture modifiers (e.g., for medications: dose, unit, frequency, route; for tests: numeric values and units; for diseases: activity status), and to infer latent modifiers implied by panel tests (e.g., default units). All machine-inferred fields are flagged as “inferred” to facilitate audit and downstream use.

2.2. Implementation and LLM configuration

CLINES is model-agnostic; we employed three instruction-finetuned frontier LLMs with identical decoding settings: Llama-3.1-405B (temperature

0.6, max output 8192 tokens; served in FP8 precision—the official Meta-Llama-3.1-405B-Instruct-FP8 checkpoint, the lightest weights released by Meta—on 8×H100 GPUs via SGLang with eight-way tensor parallelism), GPT-4o (temperature 0.6, max output 8192), and o3-mini (medium reasoning effort, max output 32768). Notes were chunked to ≤ 768 tokens of raw text per request. With GPT-4o, mean prompt tokens were ≈ 8088.25 and mean completion tokens ≈ 2795.25 per chunk; with o3-mini, mean completion tokens increased to ≈ 24334.50 . GPT-4o and o3-mini were accessed via API (no local GPUs). We distinguish the four architectural stages (chunk $\rightarrow$ extract $\rightarrow$ enrich $\rightarrow$ aggregate) from the underlying LLM invocations: a single chunk traversal issues several model calls—entity extraction, attribute and assertion classification, date normalization, and JSON-repair helper calls—so the per-note invocation count substantially exceeds four. Exact per-step invocation counts, token usage, wall-clock time, and estimated cost are reported per dataset and model in the cost analysis (Figure 5b).

All language-model inference was performed via Harvard Medical School’s HIPAA-compliant Azure OpenAI Service tenant, governed by a Business Associate Agreement between Harvard Medical School and Microsoft. Clinical text remained within the HMS-controlled Azure tenant throughout inference; no clinical data were transmitted to or stored by OpenAI directly. Microsoft Azure OpenAI Service does not retain submitted prompts or completions for model training under its service-specific terms, in contrast to the public OpenAI consumer API. This deployment configuration was reviewed for compliance with the data-use agreements of the MIMIC-III, 4CE, and CORAL datasets prior to any inference.

CLINES outputs were exported to the i2b2 schema, and a read-only web demonstrator populated with results from the credentialed-access CORAL dataset was deployed; access is described in the Data and Code availability section.

2.3. Study design and data

All corpora were de-identified and accessed under the respective data-use agreements and governance requirements of each dataset; no patient contact, intervention, or access to identifiable private information occurred. All annotators completed CITI Human Subjects Research training and held the access credentials required for each dataset prior to data access. Because this work analyzes only fully de-identified secondary data that are not readily identifiable, it does not constitute human-subjects research under the U.S. Common Rule (45 CFR 46.104) and did not require Institutional Review Board approval or a formal exemption application at the authors’ institution; further detail is provided in the Ethics Approval section.

MIMIC-III³⁰: from 2,065,096 critical-care notes, 964 notes were sampled and segmented into 2,000 paragraphs. Twenty-four paragraphs were manually annotated (one expert per paragraph), yielding 448 clinical phrases: 235 body-location mentions, 128 modifiers (e.g., assertion status), and 144 numerical values with units. Acronym–expansion pairs were treated as single phrases; narrative “purpose” attributes were excluded from quantitative analysis.

4CE²⁴: de-identified notes from patients with a history of COVID-19. Of 56 notes, 21 were annotated, producing 5,731 entity mentions with temporal information and attributes; mean document length 1,466 words (SD 636).

CORAL²: 56 de-identified oncology reports (breast or pancreatic cancer). Twenty-nine notes were annotated and partitioned into two subsets: CORAL–Breast (13 documents, 3,900 entity records; mean 1,997 words) and CORAL–Pancreas (16 documents, 4,717 entity records; mean 1,906 words). Across the annotated 4CE + CORAL subsets (14,348 entity records), 2,517 (17.5%) contained both a numerical value and a unit, and 5,680 (39.6%) carried explicit temporal information.

2.4. Annotation protocol

Five annotators followed a consensus protocol balancing accuracy and efficiency: (i) extract core entity spans while excluding modifiers to avoid redundancy; (ii) infer temporality and assertion status from surrounding context (e.g., “diagnosed with lung cancer 5 months ago,” “rule out pneumonia”); (iii) expand abbreviations and non-standard terms to full medical concepts, later normalized via SapBERT²⁷; (iv) infer missing units for numerical values. Annotation was supported by a custom React-based tool enabling interactive review, conflict resolution, and structured export. All annotators received unified training with shared examples; an independent annotator audited a sample for accuracy and consistency. Throughout, words denotes whitespace- and punctuation-delimited tokens (stopwords included; a numeric value together with its unit is counted as a single word), and a clinical phrase denotes a contiguous span corresponding to a single UMLS-mappable concept or a single value-plus-unit, as marked by annotators.

2.5. Inter-annotator agreement

To quantify the inherent variability among clinical annotators following the protocol described above, we performed a structured cross-annotation study on three notes per dataset (4CE $n = 3$; CORAL $n = 3$; total 2,657 candidate-mention rows). For each note, a second annotator, independent of the original assignment, re-annotated the same AI-suggested draft, blinded to the original annotator’s selections.

We report inter-annotator agreement (IAA) using three complementary metrics. The primary metric is the F1-based IAA (Dice coefficient), defined as $2|A \cap B|/(|A| + |B|)$, where A and B are the sets of mentions retained by each annotator. F1 is the appropriate primary IAA in span-prediction settings where annotators audit an AI-suggested draft rather than perform blank double-annotation, because the resulting label distribution is highly skewed toward keep (here $\approx 75\%$ of candidate rows kept by either annotator), conditions under which Cohen’s κ suffers from the well-documented two-paradox effect^{31,32} and systematically underestimates substantive agreement³³. We additionally report Cohen’s κ and the Prevalence- and Bias-Adjusted Kappa (PABAK)³² as supplementary measures to address Cohen’s paradox.

Annotator review uncovered a small number of recurring structural-disambiguation patterns where guideline ambiguity (rather than substantive clinical disagreement) generated apparent label divergence: (a) AI-suggested mentions whose UMLS semantic type is non-clinical (e.g., Bird, Geographic Area, Calendar Month); (b) candidate spans that match a section header (e.g., Review of Systems, Past Medical History); (c) header-like attribute types (e.g., Clinical Attribute, Body Location or Region) with Not associated or Absent asser-

tion status; (d) drug brand/generic duplicates within the same span and value/unit; and (e) standalone severity adjectives (e.g., moderate, severe). We pre-specified that under the consensus protocol all five patterns should resolve to a uniform DROP decision; the resulting reconciliation is reported alongside raw agreement.

Across the cross-annotated subset, the primary F1-based IAA was 0.837 (mention-level), with supplementary Cohen’s $\kappa = 0.597$ and PABAK = 0.609; raw agreement was 80.5%. Under conventions for F1-based IAA in clinical NLP³³, this represents substantial agreement, consistent with published clinical-NER inter-annotator agreement (e.g., i2b2 2010 challenge $\kappa = 0.43$ – 0.60 for entity-boundary tasks³⁴). Per-note metrics and the rule-trigger breakdown are reported in Supplementary Table S1. For MIMIC-III, where each paragraph was annotated by a single expert without independent re-annotation, we reframe F1 as agreement-with-annotator rather than as a ground-truth benchmark.

2.6. Baselines

To contextualize performance, we compared classical rule-/lexicon-driven pipelines and compact Transformer encoders, representing the two dominant traditions in clinical NLP.

Rule-/lexicon-based systems. cTAKES⁵ implements sectioning, sentence boundary detection, tokenization, POS tagging, morphological normalization, UMLS mapping, and ConText for negation, uncertainty, and experience; shallow parsing and dependency analysis link entities to attributes. MetaMap⁶ performs POS tagging, variant generation, composite noun-phrase formation, and UMLS matching with heuristic scoring plus statistical word-

sense disambiguation; it does not natively extract numerical values, units, or temporal relations.

Transformer encoders. Clinical-MobileBERT and Clinical-DistilBERT^{16,17} fine-tuned on i2b2-2010 concepts³⁴ were included as parameter-efficient clinical NER baselines. To address the absence of full-size deep-learning comparators, we additionally evaluated two larger, publicly released clinical NER encoders as off-the-shelf baselines (no fine-tuning on our corpora): a BERT-base clinical NER model (samrawal/bert-base-uncased_clinical-ner, 110M parameters) and a GatorTron-base clinical NER model (345M parameters, derived from the UF/NVIDIA GatorTron encoder¹⁵). Evaluations of these encoders were limited to entity extraction because they do not natively support ontology normalization, attribute extraction, or temporal reasoning. Model configurations and hyperparameters followed the authors’ implementations.

Single-prompt LLMs. Llama-3.1-405B and GPT-4o were also evaluated in a single-call setting using a compact prompt specifying the target schema and strict JSON output; notes were chunked only to satisfy context limits, and each chunk was processed once without retrieval, tools, cross-chunk aggregation, or multi-step reasoning.

2.7. Evaluation metrics and protocol

In the training-free setting (no task-specific supervised fine-tuning on the annotated test sets; the pipeline relies on prompt engineering with few-shot in-context examples, retrieval-augmented normalization, and post-processing rules, none of which constitute zero-shot learning in the strict sense), we evaluated: entity detection (phrase-level precision, recall, F1 with exact-span

match); attribute extraction (joint exact match of span and attribute for body location, assertion status, value/unit); date extraction and normalization (agreement with gold ISO dates; relative dates correct if the resolved absolute date matches the gold anchor). For Clinical-MobileBERT/Clinical-DistilBERT we report only NER; for cTAKES and MetaMap we report concept identification and assertion where available. End-to-end structuring, normalization, attribute extraction, and date handling are compared only across systems that produce those outputs. This study is reported in accordance with the MI-CLAIM³⁵ and TRIPOD+AI³⁶ reporting guidelines for artificial-intelligence studies in healthcare; completed checklists are provided in the supplementary materials, replacing the STROBE checklist of the original submission, which is intended for observational epidemiological studies rather than model development and evaluation.

3. Results

3.1. Primary performance and backbone comparison

We report CLINES with an o3-mini-medium backbone as the reference configuration, and additionally evaluate the same pipeline with GPT-4o, Llama-3.1-405B, and the smaller Phi-4 backbone. Across the three consensus-annotated corpora (4CE, CORAL–Breast, CORAL–Pancreas) and four extraction tasks, CLINES (o3-mini-medium) achieved code-level (UMLS-normalized) F1 of 0.874 (95% CI 0.849–0.896) on 4CE, 0.814 (0.787–0.838) on CORAL–Breast, and 0.848 (0.834–0.865) on CORAL–Pancreas; assertion-status F1 of 0.88 / 0.84 / 0.87; numeric-value F1 of 0.82 / 0.80 / 0.91; and mention-level F1 of 0.91 / 0.88 / 0.91 (Figure 3A–D). o3-mini-medium was the strongest

backbone overall, with CLINES (GPT-4o) trailing by ≈ 10 pp (code F1 0.78 / 0.70 / 0.78) and the open-weight CLINES (Llama-3.1-405B) trailing by ≈ 20 pp (code F1 0.67 / 0.61 / 0.66); the smaller Phi-4 backbone failed to emit normalizable codes. The o3-mini-medium configuration is therefore the reference system used throughout.

Takeaway. CLINES delivers high entity-normalization and assertion accuracy on the reference (o3-mini-medium) backbone, and all three CLINES backbones substantially outperform every non-CLINES baseline (single-prompt LLMs, deep-learning encoders, rule/lexicon systems): the architecture, not any specific backbone, drives the structured-extraction quality. Within CLINES, accuracy follows a clear hierarchy (o3-mini-medium > GPT-4o > Llama-3.1-405B), so sites trading accuracy for cost/privacy/governance can use the open-weight backbone with a quantified ≈ 20 pp code-F1 penalty.

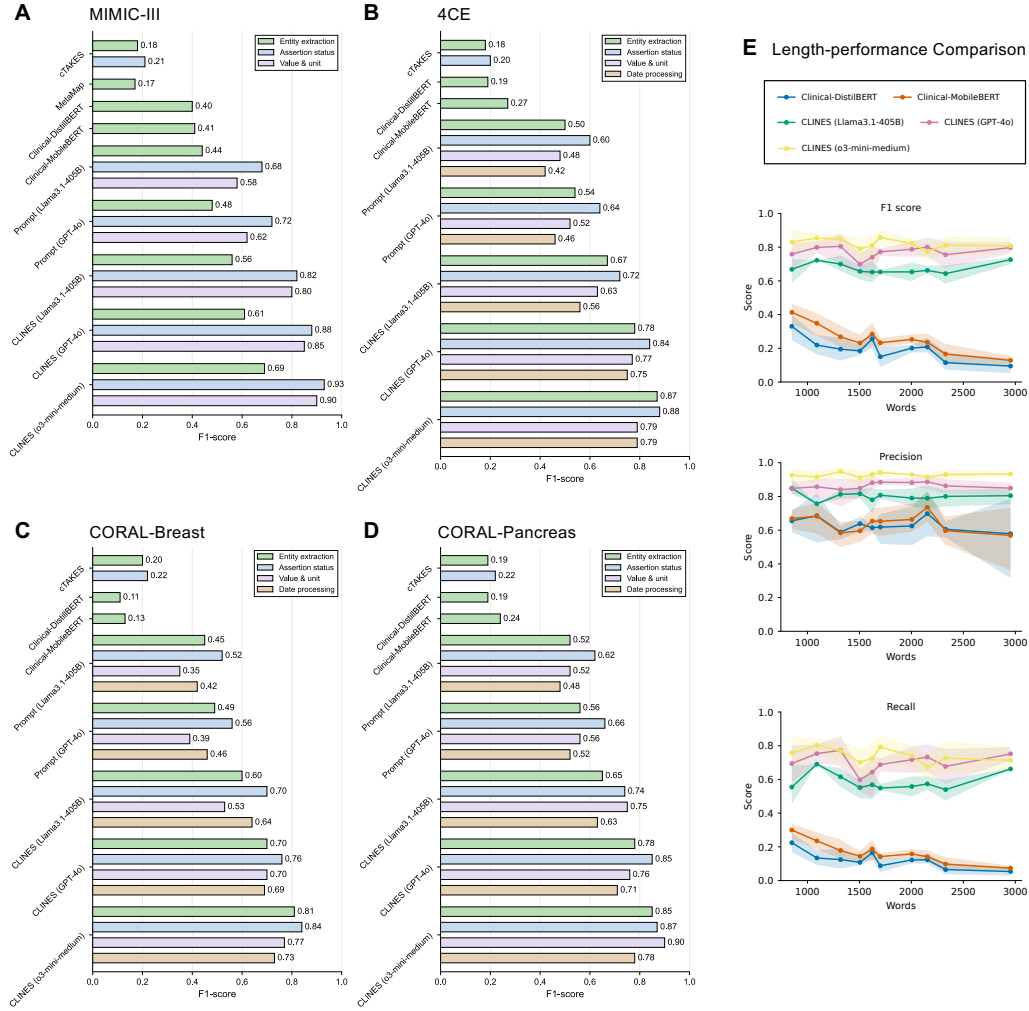


Figure 3: Performance of CLINES across LLM backbones, with 95% confidence intervals. (A–D) F1 for entity extraction, assertion status, value & unit, and date processing (where applicable) on MIMIC-III, 4CE, CORAL–Breast, and CORAL–Pancreas, for CLINES instantiated with three LLM backbones (o3-mini-medium, GPT-4o, Llama-3.1-405B) and compared with rule-based baselines (cTAKES, MetaMap), fine-tuned clinical encoders (Clinical-DistilBERT, Clinical-MobileBERT), and single-prompt LLM baselines (Prompt-Llama-3.1-405B, Prompt-GPT-4o). Error bars are 95% percentile bootstrap confidence intervals ($B = 2000$ document-level resamples). o3-mini-medium is the strongest backbone overall, with GPT-4o trailing by ≈ 10 pp co₂₁ F1 and the open-weight Llama-3.1-405B by ≈ 20 pp; differences among the strongest backbones on the date columns are not resolvable at these sample sizes. (E) F1, precision, and recall versus note length for CLINES with the three backbones and the two encoder baselines; markers indicate bin-median word counts, lines show bin-mean scores, and shaded ribbons denote 95% bootstrap confidence intervals.

3.2. Statistical confidence and stability under small evaluation sets

All primary comparisons now carry 95% percentile bootstrap confidence intervals ($B = 2000$ document-level resamples), and pairwise CLINES-versus-baseline differences were tested with a paired bootstrap and Benjamini–Hochberg false-discovery-rate correction (Supplementary Table S2). CLINES (o3-mini-medium) exceeded both backbone alternatives on code, assertion, value, and unit extraction, with absolute F1 gains spanning +0.01 to +0.21 versus GPT-4o and Llama-3.1-405B (smallest gap on 4CE Unit, largest on CORAL–Breast Code) and +0.35 to +0.87 versus Phi-4. The Code-field comparisons (largest effect sizes) and most assertion/value gains were significant at $p_{\text{FDR}} < 0.05$; the smallest per-field gaps (e.g. 4CE Unit at +0.01) fall within bootstrap noise and were not significant. On the date columns, differences among the strongest backbones were not statistically resolvable at the available sample sizes. A document-level resampling stability analysis quantified the uncertainty arising from the small annotated sets: on CORAL–Breast ($n = 13$) the bootstrap standard deviation of F1 reaches ≈ 0.04 – 0.05 , so F1 differences below ≈ 0.10 are not reliably distinguishable. We therefore rest the principal claims on the comparisons whose effect sizes exceed this threshold, and Figure 3E annotates per-bin note counts to discourage over-interpretation of sparsely populated bins.

3.3. Comparison with single-prompt and deep-learning baselines

To isolate the contribution of the multi-step architecture from that of the backbone model and the prompt style, we compared CLINES against two single-call baselines and two full-size clinical deep-learning encoders (Figure 5d). On code-level extraction, o3-mini single-prompt reached F1 0.50 /

0.43 / 0.36 (4CE / CORAL–Pancreas / CORAL–Breast) and GPT-4o chain-of-thought 0.40 / 0.34 / 0.27, versus CLINES 0.87 / 0.85 / 0.81—a gain of +0.37 to +0.54 F1 attributable to the architecture rather than to the backbone model or to chain-of-thought prompting. The deep-learning encoders (BERT-base clinical NER, 110M; GatorTron-base, 345M) were competitive on mention detection (F1 0.875 / 0.880 / 0.888 and 0.840 / 0.847 / 0.876, respectively, versus CLINES 0.906 / 0.881 / 0.913) but emit entity spans only: they perform no UMLS normalization, assertion, value/unit, or date extraction, so their code-level and attribute F1 are undefined. Classical rule- and lexicon-based systems (cTAKES, MetaMap) trailed substantially on entities; among them only cTAKES produced assertion labels, and neither produced value–unit or date outputs.

Takeaway. The multi-step architecture, not the backbone or the prompt style, drives the gains: a single call to the same model—or to a reasoning-tuned model with chain-of-thought—recovers less than half of CLINES’s normalized-extraction performance, and strong span detectors do not provide the normalized, attributed, time-stamped output that downstream cohort construction requires.

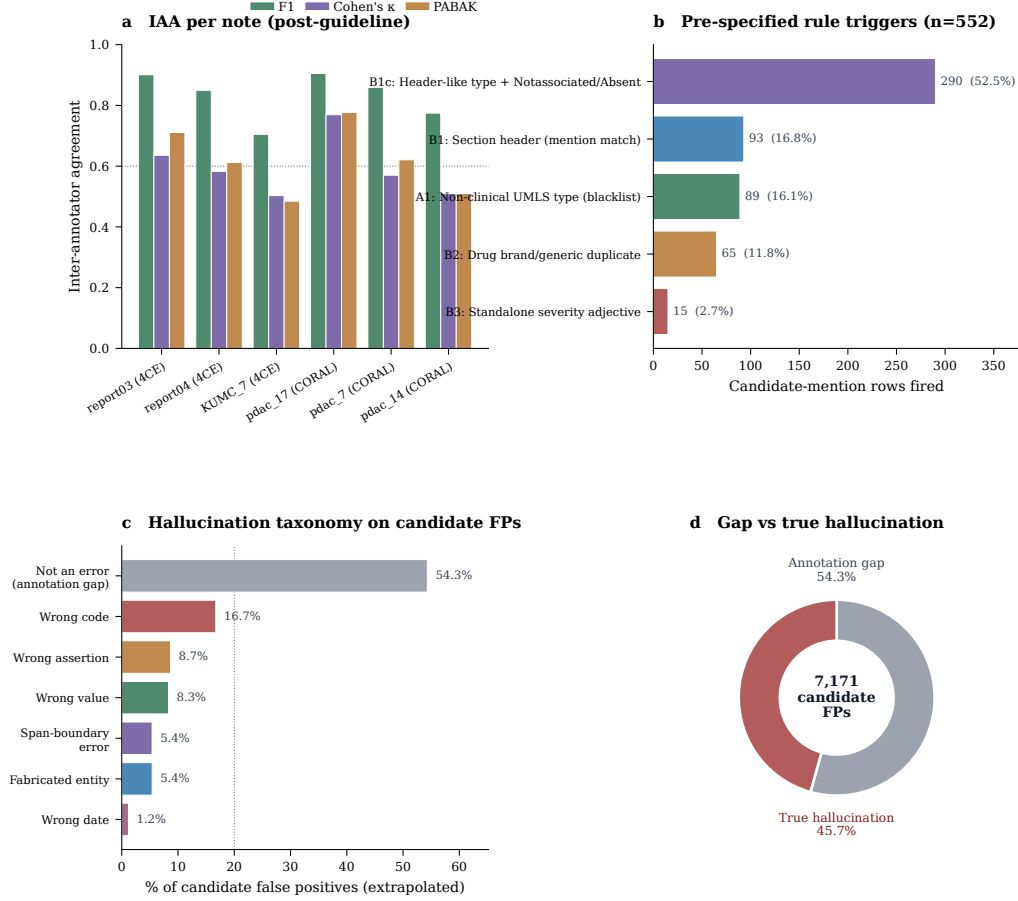


Figure 4: Inter-annotator agreement and hallucination analysis. (a) Mention-level inter-annotator agreement per cross-annotated note (F1-based IAA, Cohen’s κ , and PABAK); pooled across the cross-annotated subset, F1= 0.837, κ = 0.597, PABAK= 0.609. (b) Pre-specified structural-disambiguation rule triggers (n = 552; 15.6% of candidate-mention rows), the principal source of apparent annotator divergence. (c) Seven-class hallucination taxonomy applied to candidate false positives, with population-level rates extrapolated to the full false-positive set (N = 7,171) by post-stratified Horvitz–Thompson estimation. (d) Annotation gap versus true hallucination: 54.3% of apparent false positives are not true errors (annotation gap, synonym equivalence, or guideline ambiguity), leaving a 45.7% true-hallucination share of false positives ($\approx$ 24.4% of all predictions).

3.4. Inter-annotator agreement

On the cross-annotated subset (three notes per dataset), mention-level F1-based inter-annotator agreement was 0.837, with supplementary Cohen’s $\kappa = 0.597$ and PABAK= 0.609 (raw agreement 80.5%; Figure 4a). Under conventions for F1-based agreement in clinical NLP this represents substantial agreement, consistent with published clinical-NER inter-annotator agreement (i2b2 2010 challenge, $\kappa = 0.43$ –0.60 for entity-boundary tasks). The bulk of apparent disagreement was structural rather than substantive: five pre-specified disambiguation patterns accounted for 15.6% of candidate rows (Figure 4b). These agreement figures provide a human reference for interpreting the model-versus-gold differences reported above.

3.5. Hallucination and error analysis

We applied a seven-class taxonomy to 700 stratified false-positive predictions adjudicated by an LLM judge, with an author re-judging 60 cases (author–judge agreement 88.3%). Extrapolated to the full false-positive population ($N = 7,171$) by post-stratified Horvitz–Thompson estimation, 54.3% of apparent false positives were not true errors—annotation gaps, synonym equivalence, or guideline ambiguity—while the six true-hallucination categories together accounted for 45.7% of false positives, or $\approx 24.4\%$ of all predictions (Figure 4c–d). The most frequent true-error category was incorrect UMLS-code assignment (16.7%); outright fabricated entities (5.4%) and fabricated dates (1.2%) were rare. This quantifies the residual error that mandates human verification for safety-critical use and motivates the retrieval-grounded mitigation discussed below.

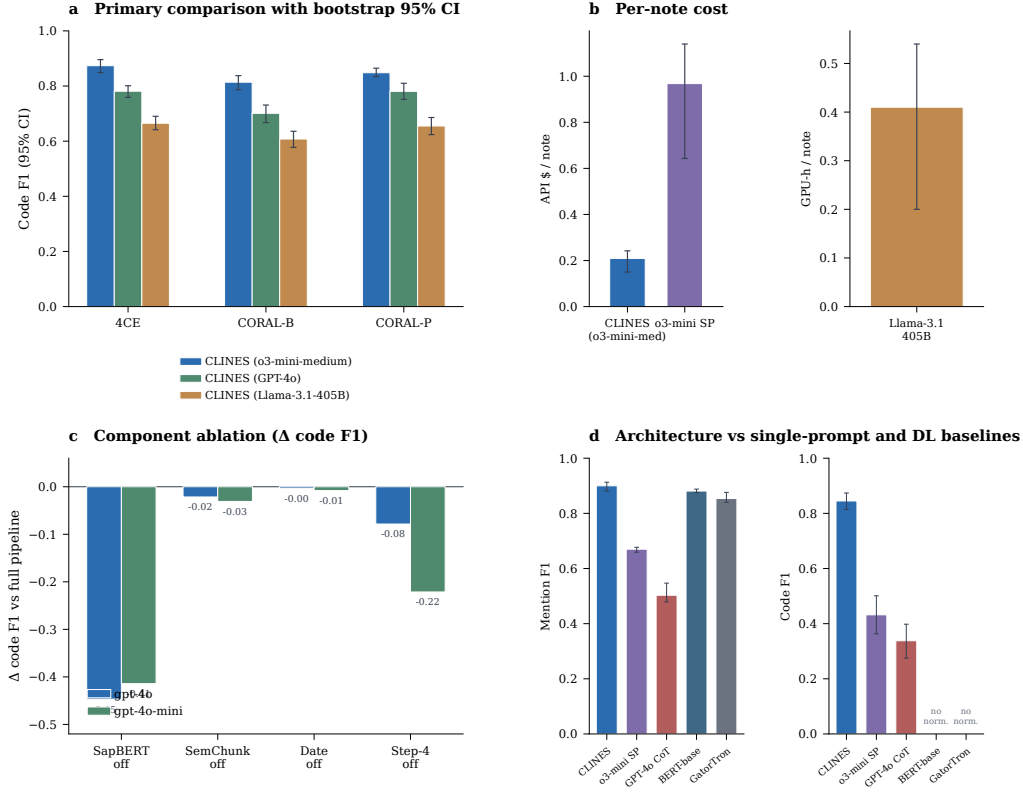


Figure 5: Robustness, cost, and component contributions. (a) Primary CLINES (o3-mini-medium) code-F1 comparison against backbone alternatives (CLINES (GPT-4o), CLINES (Llama-3.1-405B)) with 95% percentile bootstrap confidence intervals. (b) Per-note cost: hosted-API dollars per note (CLINES (o3-mini-medium) versus the same backbone in single-prompt mode) and local GPU-hours per note (Llama-3.1-405B FP8 on 8xH100); the dollar and GPU-hour axes are not directly comparable. (c) Component ablation (change in code F1 relative to the full pipeline, mean of three datasets) for two backbones (GPT-4o and GPT-4o-mini); removing SapBERT-based UMLS normalization dominates on both backbones. (d) CLINES versus single-prompt LLM baselines (o3-mini single-prompt, GPT-4o chain-of-thought) and full-size clinical deep-learning encoders (BERT-base clinical NER, GatorTron-base); the encoders emit entity spans only and produce no UMLS codes, so their code F1 is undefined.

3.6. Cost, latency, and computational efficiency

Per-note cost varied by an order of magnitude across deployment options (Figure 5b). The reference CLINES (o3-mini-medium) configuration cost approximately \$0.21 per note in hosted-API charges, versus \$0.97 per note for the o3-mini single-prompt reasoning configuration, whose long reasoning traces inflate completion-token counts even at the same backbone model. The open-weight Llama-3.1-405B (FP8, 8×H100) required ≈ 0.4 GPU-hours per note. Because each note triggers several LLM invocations per chunk, the per-note call count exceeds the four architectural stages; the full per-step invocation, token, time, and cost breakdown is provided in the supplementary materials.

3.7. Component ablation

Disabling each pipeline component in turn (Figure 5c) showed that SapBERT-based UMLS normalization is the load-bearing component: removing it reduced code-level F1 by 0.447 (GPT-4o) and 0.414 (GPT-4o-mini) on average. Cross-chunk reconciliation contributed -0.078 (GPT-4o) and -0.221 (GPT-4o-mini), semantic chunking -0.021 / -0.031 , and the date module -0.003 / -0.008 . Notably, reconciliation mattered more for the cheaper backbone, indicating that the deterministic scaffolding partly compensates for a weaker model—an argument in favor of the architecture precisely in cost-constrained settings. The ablation used two representative notes per dataset and reports point estimates.

3.8. Note-length analysis

Across quantile-based length bins ($\sim 1\text{--}3\text{k}$ words), CLINES (o3-mini-medium) maintained stable F1, precision, and recall (Figure 3E). Consistent with the original submission, lightweight transformer-encoder baselines by contrast lost recall as notes lengthened.

3.9. Performance on MIMIC-III

On the MIMIC-III subset (24 expert-annotated paragraphs), CLINES reached entity F1 0.69, assertion F1 0.93, and value+unit F1 0.90. The lower entity F1 relative to 4CE and CORAL reflects (i) the single-annotator-per-paragraph protocol used for MIMIC-III, which raises gold-standard variability and which we accordingly interpret as agreement-with-annotator rather than against a consensus reference, and (ii) the heterogeneous, abbreviation-dense nature of critical-care notes, whose ICU-specific terminology is under-represented in the few-shot examples. MIMIC-III was excluded from the bootstrap and pairwise-significance analyses above because it lacks the consensus annotation used for the other corpora.

4. Discussion

4.1. Principal findings

The CLINES pipeline advances the automation of structuring clinical narratives, transforming narrative notes into standardized data formats that can be readily ingested into relational databases or analytical data warehouses. By design, CLINES bridges the longstanding divide between narrative clinical documentation and structured queryable formats, enabling

scalable solutions for cohort discovery, real-world evidence generation, and clinical decision support. Critically, its robustness across diverse note types and lengths positions CLINES to unlock the vast reservoirs of unstructured text in EHRs. At scale, this capability can accelerate translational research, enhance population health monitoring, and power next-generation learning health systems by making rich narrative data systematically analyzable.

4.2. Backbone model considerations

Our findings reaffirm that backbone choice affects downstream performance. The finalized evaluation establishes a clear hierarchy within CLINES (reasoning-tuned o3-mini-medium leads, hosted GPT-4o trails by ≈ 10 pp code F1, and the open-weight Llama-3.1-405B trails by ≈ 20 pp), but every CLINES backbone substantially outperforms every non-CLINES baseline (single-prompt LLMs, rule/lexicon systems, deep-learning encoders); the architecture’s contribution dominates the backbone’s. Institutions can therefore make a transparent accuracy/cost/privacy trade-off: hosted reasoning models maximise accuracy where API egress is acceptable, while the open-weight backbone remains a viable lower-accuracy option for governance-constrained deployment. Comparisons across backbones should not be read as controlled head-to-head model evaluations, since the models differ in architecture, pretraining data, and token limits. A key advantage of CLINES is flexibility: the pipeline can be adapted to most LLMs so users can choose the backbone that aligns with performance, cost, and governance priorities. With the recent availability of GPT-OSS, CLINES can be straightforwardly updated to leverage such advances while maintaining accessibility.

4.3. Positioning versus existing approaches

Traditional clinical NLP—predominantly rule- or dictionary-driven—faces limited portability and high maintenance across institutions and note types; systems such as cTAKES and MetaMap emphasize lexicon/UMLS mapping and context heuristics but provide partial coverage for attributes or temporal relations. Compact transformer encoders fine-tuned for clinical NER are efficient for span detection yet generally stop short of normalization, assertion/date attribution, or cross-sentence temporal reasoning, and their recall degrades as notes lengthen. By contrast, single-prompt LLM baselines mitigate some gaps (handling heterogeneous phrasing and implicit cues) but remain brittle for schema-aligned outputs: they do not reliably perform controlled-vocabulary normalization and are more prone to hallucination without systematic designs. In CLINES, a [multi-step](#) orchestration—chunking, tool-guided extraction, normalization, and date reconciliation—delivers structured outputs with transparent intermediate artifacts and maintains robustness across diverse note types. This positions CLINES as a scalable and sustainable approach for structuring clinical narratives. [Contemporary production and research systems address overlapping goals: production clinical-NLP platforms such as John Snow Labs Healthcare NLP / Spark NLP³⁷ provide broad clinical NER, assertion, multi-terminology resolution, and OMOP output within hybrid LLM pipelines; CLEAR³⁸ is a closely related retrieval-augmented LLM pipeline for clinical extraction; and a recent multi-model evaluation³⁹ benchmarked output consistency across many LLMs. CLINES is positioned not as a competitor to mature production suites but as a transparent, fully-specified, model-agnostic open reference](#)

implementation that prioritizes reproducibility of its prompt suite and an auditable intermediate representation. We did not undertake a head-to-head re-benchmark against these systems—a fair matched comparison (identical preprocessing, gold standard, and span-matching protocol) is a substantial undertaking left to future work—and we note this as a limitation rather than a claim of superiority.

4.4. Case studies: robust clinical inference

De-identified 4CE and CORAL notes show four recurring capabilities: inferring units when measurements are underspecified; normalizing terse or site-specific shorthand to controlled vocabularies; recovering relations that mirror care processes; and handling nuanced assertion status and qualitative values. Examples include resolving “Lat 2.4,” “BE −30,” and “Glic 617” to lactate, base excess, and glucose with appropriate units and UMLS links; mapping abbreviations such as IOT, TI, RX, PCI, and EVAR to procedures or imaging; linking coronary artery disease to s/p PCI; chaining biliary obstruction → stent placement → infectious management in oncology follow-ups; and retaining Present / Absent / Possible / Conditional assertions and qualitative ranges (e.g., “mid-90s” oxygen saturation) in machine-actionable form. These vignettes show the system operating as a clinical reasoner rather than a string matcher, reducing manual normalization burden while increasing fidelity for phenotyping and decision support in heterogeneous, length-variable notes.

4.5. Dataset-level variation and generalization

Performance varies between datasets for a given model and task, reflecting different writing styles and note sizes across disease areas. The [multi-step](#), stepwise design limits error propagation: each module is independently validated, enabling interpretable error analysis. Semantic chunking is crucial for fitting long notes into LLM context windows, allowing even 4k–8k-token models to operate effectively. UMLS term normalization addresses LLM imprecision with controlled vocabularies, unifying semantically equivalent phrases (e.g., “heart attack” and “myocardial infarction”) under consistent codes and enhancing interoperability.

4.6. Schema adaptability

CLINES adapts output to target schemas (e.g., i2b2 star schema, registries, decision-support formats) and exports JSON, CSV, or SQLite, supporting integration with institutional analytics pipelines and EHR-linked databases.

4.7. Implications

For cohort discovery, phenotyping, and real-world evidence, [CLINES can substantially reduce—though not eliminate—manual chart review for entities, assertions, values/units, and dates while leaving a verifiable audit trail. Given the residual hallucination rate quantified above, we recommend that adopters retain human verification for any extraction destined for safety-critical decisions.](#) Sites can tailor prompts, dictionaries, and export schemas to local workflows without retraining large models.

4.8. Deployment considerations

Realistic deployment of CLINES depends on factors outside the extraction algorithm itself. First, input: CLINES assumes de-identified, plain-text notes, and producing that input from raw EHR sources requires PHI de-identification, format normalization, and section/document segmentation, which are institutionally heterogeneous and outside the present scope. Second, compute: the open-weight Llama-3.1-405B backbone was served in FP8 precision on 8×H100 GPUs—the only single-node configuration that accommodates a 405B model—which entails a quality trade-off relative to FP16 and an infrastructure footprint beyond the reach of many small and medium centers; the hosted-API backbones substantially lower this barrier (Figure 5b). Third, governance: the hosted-API backbones (GPT-4o, o3-mini) were run within Harvard Medical School’s HIPAA-compliant Azure OpenAI tenant under a Business Associate Agreement; this configuration does not by itself satisfy GDPR data-residency requirements outside the relevant region, so institutions with EU data-residency obligations should prefer a locally deployed open-weight backbone.

4.9. Interoperability with production terminologies

CLINES emits UMLS CUIs paired with preferred terms. UMLS is an ontology-grounded intermediate representation rather than a production interchange format: downstream use in FHIR resources or common data models requires mapping CUIs onward to the appropriate target vocabularies—SNOMED CT (via UMLS cross-source mappings), LOINC (laboratory observables), or RxNorm (medications)—and OMOP-CDM ingestion via its

standardized vocabularies, with attribute-field standardization. We therefore describe CLINES output as an ontology-grounded intermediate representation suitable for downstream mapping into production terminologies, rather than as a directly FHIR-ready artifact, and we recommend retaining a SNOMED CT intermediate layer in deployments that target FHIR.

4.10. Ethical considerations in demographic-attribute extraction

CLINES includes a prompt that can extract demographic attributes (e.g., race, ethnicity, sex, and geographic identifiers). These fields are included to support downstream cohort discovery and health-equity auditing, not model personalization. We have not evaluated whether extraction accuracy varies across demographic subgroups, and we explicitly recommend a subgroup-stratified evaluation as a required pre-deployment step for any institutional adopter. Downstream users bear responsibility for applying appropriate access controls when demographic fields are stored or analyzed.

4.11. Limitations

Several limitations remain. (i) Prompt engineering influences performance at module level; prompts were not dataset- or model-optimized. (ii) Comparisons among o3-mini/GPT-4o and Llama-3.1-405B are not apples-to-apples given architecture, data, and token-limit differences. (iii) Hallucination risk persists, especially for ambiguous references or rare abbreviations (e.g., “OD 2 gtt qd” misread as overdose rather than ophthalmic dosing). We characterized this risk by manually adjudicating a stratified sample of candidate false-positive predictions under a seven-class hallucination taxonomy (fabricated entity, span-boundary error, wrong assertion, wrong code, wrong

value, wrong date, not-an-error). The dominant category—accounting for the majority of disagreements with single-annotator gold—was not-an-error, indicating that a large fraction of apparent false positives reflect annotator omission or guideline ambiguity rather than model fabrication; per-category rates of true hallucination categories were each below 17% of audited cases. (iv) Gold annotations came from different annotators across datasets, introducing variability. We quantified this variability through a structured cross-annotation study on three notes per dataset ($n = 6$ notes, 2,657 candidate-mention rows), establishing a mention-level F1-based inter-annotator agreement of 0.837 (Cohen’s $\kappa = 0.597$; PABAK = 0.609) under the consensus protocol described in Methods. Differences between model predictions and the single-annotator gold standard that fall within these human-agreement bounds should be interpreted as within the inherent variability of expert annotation, not as model error. For datasets annotated by a single expert per paragraph (notably MIMIC-III), F1 scores are accordingly reframed as agreement-with-annotator references rather than absolute ground-truth benchmarks.

For each limitation we note a path forward: prompt sensitivity (i) can be addressed with automatic prompt-search and optimization workflows; residual hallucination (iii) can be reduced with retrieval-grounded extraction that cites source spans and thereby enables provenance checking; and annotation variability (iv) can be narrowed by extending double-annotation to a larger validation cohort. Three further limitations bound the present claims. (v) The evaluation covers critical-care, COVID-era, and oncology notes but not several common note types (discharge summaries, operative and radiology

reports), so generalizability beyond the evaluated distributions is untested; the bootstrap and significance analyses were further limited to the three consensus-annotated corpora. (vi) The relation-extraction module, though deployed, is not benchmarked here because gold-standard relation annotations are unavailable. (vii) We did not exhaustively optimize the single-prompt and chain-of-thought baselines, in order to avoid an unfair comparison in either direction; their reported performance should therefore be read as a reasonable rather than a maximally-tuned reference.

4.12. Future work

Future directions include domain-adaptive fine-tuning (e.g., oncology, pediatrics), adaptive feedback loops where clinician corrections refine prompts or parameters, and multilingual extensions to support non-English health-care systems. Additional directions motivated by this revision include: hybrid per-stage model routing, in which a lightweight model performs entity extraction while a stronger model is reserved for reasoning-intensive steps (the modular design already supports per-stage backbone selection); a formal output-consistency evaluation across repeated runs, given the stochastic decoding used here; retrieval-grounded extraction with span-level provenance to mitigate hallucination; subgroup-stratified fairness evaluation of demographic extraction; benchmarking of the relation-extraction module once gold relation annotations become available; expansion to additional note types (discharge summaries, operative and radiology reports); and tighter integration with OMOP-CDM and FHIR via SNOMED CT, RxNorm, and LOINC mapping.

4.13. Conclusion

CLINES translates free-text clinical notes into ontology-grounded, attributed, time-stamped, and auditable structured data through a transparent multi-step LLM pipeline. Across three consensus-annotated corpora it substantially outperformed single-prompt and chain-of-thought LLMs and full-size clinical encoders on normalized extraction, with gains that were robust across open- and closed-weight backbones and supported by bootstrap confidence intervals and paired-bootstrap testing. A structured inter-annotator-agreement study (mention-level F1-based IAA 0.837) provides a human reference for interpreting model-versus-gold differences, and a seven-class error analysis shows that most apparent false positives are annotation gaps rather than fabrications, with a quantified residual hallucination rate that mandates human verification for safety-critical use. CLINES offers a practical, model-agnostic, and reproducible route to scale chart-review-like extraction for cohort discovery and real-world evidence, while making explicit the conditions—input preprocessing, compute, governance, and downstream terminology mapping—required for responsible deployment.

Contributors

Conceptualization & supervision: Tianxi Cai, Isaac Kohane. Methodology & software: Zongxin Yang, Hongyi Yuan, Raheel Sayeed, Amelia Li Min Tan (design); Zongxin Yang, Hongyi Yuan (implementation). Data curation & annotation: Enci Cai, Mohammed Moro, Xiudi Li, Zongxin Yang, Raheel Sayeed. [Investigation & cross-annotation \(for inter-annotator agreement\)](#): [Enci Cai, Mohammed Moro](#). Formal analysis: Zongxin Yang, Hongyi Yuan,

Huaiyuan Ying. Data access & verification: Zongxin Yang, Raheel Sayeed. The i2b2 web demonstrator: Nicholas Brown, Griffin Weber. Writing—original draft: Zongxin Yang, Raheel Sayeed. Writing—review & editing: all authors. Accountability: all authors approved the final manuscript, and accept responsibility for its accuracy and integrity. The corresponding author had final responsibility for the decision to submit.

Data Sharing

Source clinical notes were accessed under data-use agreements and are not publicly shareable. The MIMIC-III dataset is available from PhysioNet (<https://physionet.org/content/mimiciii/>) and the CORAL dataset is available from PhysioNet (<https://physionet.org/content/curated-oncology-reports/>), both under the PhysioNet credentialed access process requiring completion of CITI Human Subjects Research training and signing of the data-specific Data Use Agreement. The 4CE Phase 1.2 dataset is available through the 4CE Consortium (<https://covidclinical.net>) upon completion of the consortium data-use agreement and approval of the site data-sharing committee. Cross-annotation files used for the inter-annotator agreement analysis (de-identified mention selections on six notes) are available from the corresponding author on reasonable request, subject to verification of the requestor’s credentials for the source dataset. No additional patient-level data were generated by this study. Study-derived prompt templates and a public inference reference implementation of CLINES (single-note inference with user-supplied Azure OpenAI credentials) are released under the MIT License at <https://github.com/z-x-yang/clines>. An interactive, read-

only i2b2 web demonstrator populated with CORAL results is available at celehs.hms.harvard.edu/i2b2_demo.

Ethics Approval

This study involved secondary analysis of fully de-identified electronic health record data. Because the analysis involves only fully de-identified secondary data that are not readily identifiable, it does not constitute human-subjects research under the U.S. Common Rule (45 CFR 46.104) and did not require Institutional Review Board approval or a formal exemption determination at the authors' institution (Harvard Medical School); no patient contact, intervention, or access to identifiable private information occurred. The datasets were obtained and used under their respective data-use agreements: PhysioNet credentialed access for MIMIC-III, the CORAL data-use agreement, and 4CE consortium credentialing. All annotators completed CITI Human Subjects Research training and held the access credentials required for each dataset prior to data access. All large-language-model inference on clinical text was performed within Harvard Medical School's HIPAA-compliant Microsoft Azure OpenAI tenant under a Business Associate Agreement, with no clinical text sent to or retained by external model providers. Ethical approvals and data-use agreements for the original data collection were obtained by the respective data custodians.

Patient and Public Involvement

This study used previously collected, fully de-identified electronic health record datasets and did not involve prospective recruitment or direct con-

tact with patients. Patients and members of the public were therefore not involved in setting the research question, in the design, conduct, or interpretation of the study, or in the preparation of this manuscript. As the study analyses only de-identified secondary data, individual patients could not be identified and dissemination of results to study participants is not applicable. Future translational deployment of CLINES in care settings would benefit from patient and public involvement to inform acceptable use, transparency, and oversight, as noted in the Discussion.

Declaration of Interests

We declare no competing interests.

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References

- [1] Tayefi M, Ngo P, Chomutare T, Dalianis H, Salvi E, Budrionis A, et al. Challenges and opportunities beyond structured data in analysis of electronic health records. *WIREs Computational Statistics*. 2021 Feb;13(6). Available from: <http://dx.doi.org/10.1002/wics.1549>.
- [2] Sushil M, Kennedy VE, Mandair D, Miao BY, Zack T, Butte AJ. CORAL: Expert-Curated Oncology Reports to Advance Language Model Inference. *NEJM AI*. 2024 Mar;1(4). Available from: <http://dx.doi.org/10.1056/aidbp2300110>.

- [3] Kohane IS, Aronow BJ, Avillach P, Beaulieu-Jones BK, Bellazzi R, Bradford RL, et al. What Every Reader Should Know About Studies Using Electronic Health Record Data but May Be Afraid to Ask. *Journal of Medical Internet Research*. 2021 Mar;23(3):e22219. Available from: <http://dx.doi.org/10.2196/22219>.
- [4] Kim MK, Rouphael C, McMichael J, Welch N, Dasarathy S. Challenges in and Opportunities for Electronic Health Record-Based Data Analysis and Interpretation. *Gut and Liver*. 2023 Oct;18(2):201–208. Available from: <http://dx.doi.org/10.5009/gnl230272>.
- [5] Savova GK, Masanz JJ, Ogren PV, Zheng J, Sohn S, Kipper-Schuler KC, et al. Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications. *Journal of the American Medical Informatics Association*. 2010 Sep;17(5):507–513. Available from: <http://dx.doi.org/10.1136/jamia.2009.001560>.
- [6] Aronson AR. Effective mapping of biomedical text to the UMLS Metathesaurus: the MetaMap program. In: *Proceedings of the AMIA Symposium*; 2001. p. 17.
- [7] Friedman C, Hripcsak G, DuMouchel W, Johnson SB, Clayton PD. Natural language processing in an operational clinical information system. *Natural Language Engineering*. 1995 Mar;1(1):83–108. Available from: <http://dx.doi.org/10.1017/S1351324900000061>.
- [8] Pathak J, Kho AN, Denny JC. Electronic health records-driven phe-

- notyping: challenges, recent advances, and perspectives. *Journal of the American Medical Informatics Association*. 2013 Dec;20(e2):e206–e211. Available from: <http://dx.doi.org/10.1136/amiajnl-2013-002428>.
- [9] Lample G, Ballesteros M, Subramanian S, Kawakami K, Dyer C. Neural Architectures for Named Entity Recognition. In: *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*; 2016. p. 260-70.
 - [10] Lee J, Yoon W, Kim S, Kim D, Kim S, So CH, et al. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*. 2020;36(4):1234-40.
 - [11] Alsentzer E, Murphy J, Boag W, Weng WH, Jindi D, Naumann T, et al. Publicly Available Clinical BERT Embeddings. In: *Proceedings of the 2nd Clinical Natural Language Processing Workshop*; 2019. p. 72-8.
 - [12] OpenAI, Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, et al.. GPT-4 Technical Report. *arXiv*; 2023. Available from: <https://arxiv.org/abs/2303.08774>.
 - [13] Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, et al. Large language models encode clinical knowledge. *Nature*. 2023 Jul;620(7972):172–180. Available from: <http://dx.doi.org/10.1038/s41586-023-06291-2>.
 - [14] Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, et al..

- Language Models are Few-Shot Learners. arXiv; 2020. Available from: <https://arxiv.org/abs/2005.14165>.
- [15] Yang X, Chen A, PourNejatian N, Shin HC, Smith KE, Parisien C, et al.. GatorTron: A Large Clinical Language Model to Unlock Patient Information from Unstructured Electronic Health Records. arXiv; 2022. Available from: <https://arxiv.org/abs/2203.03540>.
- [16] Rohanian O, Nouriborji M, Kouchaki S, Clifton DA. On the effectiveness of compact biomedical transformers. *Bioinformatics*. 2023 Feb;39(3). Available from: <http://dx.doi.org/10.1093/bioinformatics/btad103>.
- [17] Rohanian O, Nouriborji M, Jauncey H, Kouchaki S, Nooralahzadeh F, Clifton L, et al. Lightweight transformers for clinical natural language processing. *Natural Language Engineering*. 2024 Jan;30(5):887–914. Available from: <http://dx.doi.org/10.1017/S1351324923000542>.
- [18] Kohane IS, Churchill SE, Murphy SN. A translational engine at the national scale: informatics for integrating biology and the bedside. *Journal of the American Medical Informatics Association*. 2012 Mar;19(2):181–185. Available from: <http://dx.doi.org/10.1136/amiajnl-2011-000492>.
- [19] Radford A, Narasimhan K, Salimans T, Sutskever I. Improving Language Understanding by Generative Pre-Training. OpenAI; 2018. Available from: https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf.
- [20] Karabacak M, Margetis K. Embracing Large Language Models for Med-

- ical Applications: Opportunities and Challenges. Cureus. 2023 May. Available from: <http://dx.doi.org/10.7759/cureus.39305>.
- [21] Rawte V, Sheth A, Das A. A Survey of Hallucination in Large Foundation Models. arXiv; 2023. Available from: <https://arxiv.org/abs/2309.05922>.
 - [22] Zhao WX, Zhou K, Li J, Tang T, Wang X, Hou Y, et al. A Survey of Large Language Models. arXiv preprint arXiv:2303.18223. 2023. Available from: <https://arxiv.org/abs/2303.18223>.
 - [23] Grattafiori A, Dubey A, Jauhri A, Pandey A, Kadian A, Al-Dahle A, et al. The Llama 3 Herd of Models. arXiv preprint arXiv:2407.21783. 2024. Available from: <https://arxiv.org/abs/2407.21783>.
 - [24] Brat GA, Weber GM, Gehlenborg N, Avillach P, Palmer NP, Chiovato L, et al. International electronic health record-derived COVID-19 clinical course profiles: the 4CE consortium. npj Digital Medicine. 2020 Aug;3(1). Available from: <http://dx.doi.org/10.1038/s41746-020-00308-0>.
 - [25] OpenAI. tiktoken. GitHub; 2024. GitHub repository, release 0.6.0 (published 2024-02-09). Available from: <https://github.com/openai/tiktoken>.
 - [26] Isaacus-dev. semchunk. GitHub; 2024. GitHub repository, version 2.0.0 (released 2024-06-19). Available from: <https://github.com/isaacus-dev/semchunk>.

- [27] Liu F, Shareghi E, Meng Z, Basaldella M, Collier N. Self-alignment Pretraining for Biomedical Entity Representations. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; 2021. p. 4228-38.
- [28] He T, Belouali A, Patricoski J, Lehmann H, Ball R, Anagnostou V, et al. Trends and opportunities in computable clinical phenotyping: A scoping review. *Journal of Biomedical Informatics*. 2023 Apr;140:104335. Available from: <http://dx.doi.org/10.1016/j.jbi.2023.104335>.
- [29] Hou J, Zhao R, Gronsbell J, Lin Y, Bonzel CL, Zeng Q, et al. Generate Analysis-Ready Data for Real-world Evidence: Tutorial for Harnessing Electronic Health Records With Advanced Informatic Technologies. *Journal of Medical Internet Research*. 2023 May;25:e45662. Available from: <http://dx.doi.org/10.2196/45662>.
- [30] Johnson AEW, Pollard TJ, Shen L, Lehman LwH, Feng M, Ghassemi M, et al. MIMIC-III, a freely accessible critical care database. *Scientific Data*. 2016 May;3(1). Available from: <http://dx.doi.org/10.1038/sdata.2016.35>.
- [31] Feinstein AR, Cicchetti DV. High agreement but low kappa: I. The problems of two paradoxes. *Journal of Clinical Epidemiology*. 1990;43(6):543-9.
- [32] Byrt T, Bishop J, Carlin JB. Bias, prevalence and kappa. *Journal of Clinical Epidemiology*. 1993;46(5):423-9.

- [33] Hripcsak G, Rothschild AS. Agreement, the F-measure, and reliability in information retrieval. *Journal of the American Medical Informatics Association*. 2005;12(3):296-8.
- [34] Uzuner Ö, South BR, Shen S, DuVall SL. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *Journal of the American Medical Informatics Association*. 2011;18(5):552-6.
- [35] Norgeot B, Quer G, Beaulieu-Jones BK, Torkamani A, Dias R, Gianfrancesco M, et al. Minimum information about clinical artificial intelligence modeling: the MI-CLAIM checklist. *Nature Medicine*. 2020;26(9):1320-4.
- [36] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*. 2024;385:e078378.
- [37] John Snow Labs. Healthcare NLP (Spark NLP for Healthcare), version 6.3; 2026. Production-grade clinical NLP platform; academic free licence. <https://www.johnsnowlabs.com/healthcare-nlp/>.
- [38] Lopez I, Swaminathan A, Vedula K, et al. Clinical entity augmented retrieval for clinical information extraction. *npj Digital Medicine*. 2025;8:45.
- [39] Ntinopoulos V, et al. Large language models for data extraction from unstructured and semi-structured electronic health records: a multi-

ple model performance evaluation. *BMJ Health & Care Informatics*.
2025;32:e101139.

Supplementary Material for CLINES

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1. Overview

This supplementary document expands the methodological details for CLINES. We provide the complete prompting strategy, clarify the structured outputs in both JSON and i2b2 formats, and outline the retrieval procedure used for mapping clinical mentions to UMLS concepts. Terminology and task framing match the main manuscript to avoid discrepancies.

2. Prompt Suite

All prompts are delivered verbatim to the instruction-finetuned models. Listings wrap lines automatically to prevent overflow. [The few-shot examples reproduced below are the exact in-context demonstrations embedded in the deployed prompt templates \(the {note} placeholder marks where the input note is inserted at run time\). They are reproduced here in full for reproducibility.](#)

2.1. Clinical Entity Extraction

Purpose. Detects problems, findings, procedures, medications, tests, and other biomedical entities; values for panel tests are separate entities.

```
## Extract biomedical entities from the electronic health record below. Include
    tests, procedures, symptoms, diseases, allergies, medications, etc. Mark
    each entity with <KEY> tags (e.g., <1>entity</1>, <2>entity</2>).
```

```
## Rules:
```

- Generally DO NOT mark values/units/routes/frequencies associated with primary entities
- EXCEPTION: For panel tests (Chem-7, CMP, CBC), mark each value as separate entity
- Include the main entity name even with abbreviations
- For overlapping entities, prioritize the most specific concept
- Mark negated findings (e.g., "no fever" -> mark "fever")

```
## Examples:
```

```
Input:
```

```
...
```

```
The patient was taken to the Operating Room for wound exploration directly from
the Trauma Room. TUMOR SIZE: 4 x 3.2 x 3 cm, Chem-7: 127, 3.6, 88, 29, 16.5,
0.6, 143.
```

```
...
```

```
Output:
```

```
...
```

```
The patient was taken to the <1>Operating Room</1> for <2>wound exploration</2>
directly from the <3>Trauma Room</3>. <4>TUMOR SIZE</4>: 4 x 3.2 x 3 cm, <5>
Chem-7</5>: <6>127</6>, <7>3.6</7>, <8>88</8>, <9>29</9>, <10>16.5</10>,
<11>0.6</11>, <12>143</12>.
```

```
...
```

Input:

Hydrocodone 7.5mg + Apap 325mg 7.5-325MG take 1 PO PRN arthritis; Pt c/o SOB and CP. EKG showed NSR. CBC with WNL RBC but WBC 12.4.

Output:

<1>Hydrocodone</1> 7.5mg + <2>Apap</2> 325mg 7.5-325MG take 1 PO PRN <3>arthritis </3>; <4>Pt</4> <5>c/o</5> <6>SOB</6> and <7>CP</7>. <8>EKG</8> showed <9>NSR</9>. <10>CBC</10> with <11>WNL</11> <12>RBC</12> but <13>WBC</13> 12.4.

Input:

{note}

Please respond with only the annotated record. Do not include any additional text
. Output:

2.2. Entity Relationship Linking

Purpose. Links nearby entities with directed clinical relations (treatment, causal, measurement, anatomical, temporal, or general relatedness).

Identify relationships between marked entities in this health record. Find which entities are clinically related to each other and determine their relationship type.

Output a JSON list of dictionaries with:

- tag: entity order number (from KEY tags)
- related: a dictionary where keys are tags of related entities and values are relationship types FROM the perspective of the entity with the "tag" key TO the entity in the "related" dictionary keys

IMPORTANT RULES:

1. Relationship Direction:

- The relationship is always FROM the entity with the "tag" field TO the entity in the "related" field keys
- For example: {"tag": "1", "related": {"2": "treated_by"}} means "Entity 1 is treated by Entity 2"
- Similarly: {"tag": "2", "related": {"1": "treats"}} means "Entity 2 treats Entity 1"

2. Proximity:

- Only identify relationships between entities that are close to each other in the text
- Focus on entities within the same sentence or adjacent sentences
- Do not try to link entities that are far apart (e.g., different paragraphs)
- Prioritize obvious, direct relationships over tenuous connections

Relationship types include (non-exhaustive):

- Treatment: treats / treated_by / manages / managed_by / alleviates / alleviated_by
- Causal: causes / caused_by / exacerbates / exacerbated_by / indicates / indicated_by / risk_factor / at_risk_from / complication_of / has_complication

- Measurement: measures / measured_by / evaluates / evaluated_by / diagnoses / diagnosed_by / monitors / monitored_by / has_value / value_of / has_unit / unit_for
- Anatomical: located_in / location_of / part_of / has_part / administered_at / site_for
- Temporal: precedes / follows / concurrent_with
- Clinical: symptom_of / has_symptom / finding_of / has_finding / manifestation_of / has_manifestation / has_dosage / dosage_for / has_frequency / frequency_for / has_route / route_for / component_of / has_component
- Default: related_to

Examples:

Input:
...

Patient with <1>hypertension</1> takes <2>lisinopril</2> 20mg daily. <3>Blood pressure</3> was <4>142/88</4> <5>mmHg</5>.

...

Output:
...

```
[
  {"tag": "1", "related": {"2": "treated_by", "3": "measured_by"}},
  {"tag": "2", "related": {"1": "treats"}},
  {"tag": "3", "related": {"1": "measures", "4": "has_value", "5": "has_unit"}},
  {"tag": "4", "related": {"3": "value_of", "5": "measured_in"}},
  {"tag": "5", "related": {"3": "unit_of", "4": "unit_for"}}
]
```

...

Input:
...

{note}
...

Please respond with valid JSON only, no additional text. Output:

2.3. Entity Normalization

Purpose. Converts abbreviations and informal mentions to standardized biomedical terms for ontology mapping.

Recover standard names/synonyms for marked entities in this health record. For abbreviations, provide full forms. For values (numbers/units), infer the biomedical concept.

Output a JSON list of dictionaries with:

- TAG: the order number of the entity (from KEY tags)
- CLEAN: standardized name/synonym for the entity

Handling ambiguity:

- Use context to disambiguate abbreviations
- For unclear abbreviations, provide most common medical expansion
- For values, infer relevant biomedical concepts
- Standardize vague symptoms to specific medical terms

```

## Example:
Input:
...
The patient reported <1>HTN</1> history and underwent <2>BP Measurement</2>,
which showed <3>150/90 mmHg</3>. Labs revealed <4>Elevated A1C</4> at
<5>7.5%</5>. Follow-up tests ruled out <6>CAD</6>, but a <7>1.5 cm Mass</7>
was detected.
...

Output:
...
[
  {"TAG": "1", "CLEAN": "Hypertension"},
  {"TAG": "2", "CLEAN": "Blood Pressure Measurement"},
  {"TAG": "3", "CLEAN": "Hypertensive Blood Pressure"},
  {"TAG": "4", "CLEAN": "Increased Hemoglobin A1C Level"},
  {"TAG": "5", "CLEAN": "Glycated Hemoglobin Measurement"},
  {"TAG": "6", "CLEAN": "Coronary Artery Disease"},
  {"TAG": "7", "CLEAN": "Mass"}
]
...

Input:
...
{note}
...

```

Please respond with valid JSON only, no additional text. Output:

2.4. Assertion Status Classification

Purpose. Assigns clinical assertion categories (Present, Absent, Historical, Possible, Conditional, Hypothetical, Notassociated) to each entity.

Determine the assertion status for every entity marked by <KEY> and </KEY> in this health record.

Output a JSON list of dictionaries with:

- tag: entity order number (from KEY tags)
- assertion_status: one of Present, Absent, Historical, Possible, Conditional, Hypothetical, Notassociated

Ambiguity guidelines:

- Prioritize most severe/current mention
- Medications: Present (active), Historical (discontinued), Conditional (PRN), Absent (refused/contraindicated), Possible (under consideration)
- "Patient at risk for X" = Hypothetical
- "No evidence of X" = Absent
- "Cannot rule out X" = Possible
- Family history -> Notassociated
- "Presenting with a history of X" or "X-day history of Y" -> treat Y as Present unless explicitly resolved

```

## Example:
Input:
...

```

CT showed <1>lesions</1> most likely secondary to <2>metastatic disease</2>. <3> Ativan</3> 0.5 mg IV q 4 to 6 hours prn <4>anxiety</4>. Patient presenting with a 7 day history of <5>headaches</5> and <6>fever</6>.

Output:

```

[
  {"tag": "1", "assertion_status": "Present"},
  {"tag": "2", "assertion_status": "Possible"},
  {"tag": "3", "assertion_status": "Present"},
  {"tag": "4", "assertion_status": "Conditional"},
  {"tag": "5", "assertion_status": "Present"},
  {"tag": "6", "assertion_status": "Present"}
]

```

Example (Medications):

Input: Patient was <1>prescribed metformin</1> but <2>refused due to insurance </2>. <3>Lisinopril</3> discontinued last month. <4>Tylenol</4> as needed for pain.

Output:

```

[
  {"tag": "1", "assertion_status": "Present"},
  {"tag": "2", "assertion_status": "Absent"},
  {"tag": "3", "assertion_status": "Historical"},
  {"tag": "4", "assertion_status": "Conditional"}
]

```

Input:

```

{note}

```

Please respond with valid JSON only, no additional text. Output:

2.5. Attribute Enrichment

Purpose. Captures values, units, administration route, frequency, anatomical site, and descriptive modifiers for each entity.

Extract detailed information for each marked entity in this health record.

Output JSON as a list of dictionaries with these keys:

- tag: entity's order number (matching the KEY number in the text)
- body_location: anatomical location related to the entity (omit if not applicable)
- value: numerical or text value(s) linked to the entity (use array for multiple values; exclude time information)
- unit: units aligned with value (array if multiple)
- infer: boolean indicating if each unit was inferred (array if multiple)
- note: for numeric values, 'greater'/'lower'/'equal' comparator (array if multiple)
- route: medication administration route (omit if not applicable)

```

- freq: medication frequency (omit if not applicable)
- other: descriptive modifiers not captured elsewhere

## Value extraction rules:
- Names/identifiers have no value
- Measurements and qualitative lab results have value
- Medication + dose: drug name has no value; dose is captured separately
- Use arrays when multiple values occur; arrays must align in length
- Time information belongs in "other", not in "value"

## Special handling:
- Panel tests: each value is a separate entity
- Infer units only when context is clear
- Measurements like "2 cm mass": value=2, unit=cm

## Examples:
Input:
...
<1>Metoprolol</1> 50 mg PO BID for <2>hypertension</2>. <3>Chem-7</3>:
    <4>127</4>, <5>3.6</5>, <6>88</6>.
...

Output:
...
[
  {"tag": "1", "value": "50", "unit": "mg", "infer": false, "note": "equal", "
    route": "PO", "freq": "BID", "other": "for hypertension"},
  {"tag": "2", "value": null},
  {"tag": "3", "value": null},
  {"tag": "4", "value": "127", "unit": "mEq/L", "infer": true, "note": "equal"},
  {"tag": "5", "value": "3.6", "unit": "mEq/L", "infer": true, "note": "equal"},
  {"tag": "6", "value": "88", "unit": "mEq/L", "infer": true, "note": "equal"}
]
...

Input:
...
<1>Glucose</1> readings: 112 mg/dL (fasting), 145 mg/dL (after meal). <2>Pain</2>
    in lower back rated 8/10 in morning, 6/10 in evening.
...

Output:
...
[
  {"tag": "1", "value": ["112", "145"], "unit": ["mg/dL", "mg/dL"], "infer": [
    false, false], "note": ["equal", "equal"], "other": "first reading was
    fasting, second was after meal"},
  {"tag": "2", "value": ["8/10", "6/10"], "body_location": "lower back", "other":
    "8/10 in morning, 6/10 in evening"}
]
...

Input:
...
<1>Irregular somewhat scalloped appearing hypoechoic focus</1> (2 cm) in left
    axillary tail. <2>Mass</2> measuring 3.5 x 2.8 cm noted in right breast.
...

```

```

Output:
...
[
  {"tag": "1", "value": "2", "unit": "cm", "body_location": "left axillary tail",
   "other": "Irregular somewhat scalloped appearing", "infer": false},
  {"tag": "2", "value": "3.5 x 2.8", "unit": "cm", "body_location": "right breast",
   "infer": false}
]
...

Input:
...
<1>Blood pressure</1> elevated at 160/90 mmHg. <2>Temperature</2> normal, <3>
pulse</3> irregular at 95 bpm.
...

Output:
...
[
  {"tag": "1", "value": ["elevated", "160/90"], "unit": [null, "mmHg"], "infer":
   [false, false], "note": [null, "equal"]},
  {"tag": "2", "value": "normal"},
  {"tag": "3", "value": "95", "unit": "bpm", "infer": false, "note": "equal", "
   other": "irregular"}
]
...

Input:
...
{note}
...

```

Please respond with valid JSON only, no additional text. Output:

2.6. Basic Patient and Visit Information

Purpose. Derives admission/discharge dates and demographics for temporal anchoring and schema export.

```

## Extract the following basic patient information from this health record:
- admission_date
- discharge_date
- gender
- death_date
- birth_date
- race
- ethnicity
- zip_code

```

Output as JSON Dictionary using the above as keys.
 For dates, use "YYYY-MM-DD" format. If information is not explicitly mentioned,
 use null.

```

## IMPORTANT DATE EXTRACTION RULES:
- "Admission Date"/"Admitted on" -> admission_date
- "Discharge Date"/"Discharged on" -> discharge_date

```

- Visit/Encounter/Appointment or Date of Service -> use as both admission_date and discharge_date if same-day visit is likely
- "Date:", "Date of Note", provider-titled dates -> use as both admission_date and discharge_date
- Only one date present -> use it for both admission_date and discharge_date

Input:
 ...
 {note}
 ...

Please respond with valid JSON only, no additional text. Output:

2.7. Per-Chunk Date Extraction (Single-chunk Context)

Purpose. Implements the date processing described in Step 3 of the Methods, assigning start and end dates to each entity using the current note as the temporal anchor.

```
## Extract dates for ALL entities marked by <KEY> tags in this health record.
    Every entity must have a date entry.

## Output JSON list of dictionaries with:
- tag: entity order number from KEY tags
- date: [YYYY-MM-DD, YYYY-MM-DD] for start and end (same date if single event;
    null if unknown)
- inferred: [true/false, true/false] indicating whether each boundary was
    inferred

## Key Rules:
- Extract visit date as reference
- Explicit dates -> exact start/end, inferred=false
- "X days/weeks/months/years ago" -> END = visit date, START = visit date minus X
    , inferred=true
- Current symptoms/findings -> visit date for both boundaries, inferred=false
- New treatments/prescriptions -> START = visit date, END = null, inferred=[false
    , true]
- Chronic conditions without timeframe -> [null, null], inferred=[false, false]
- Related events share the main event's window
- Unknown timing -> [null, null], inferred=[false, false]
```

Input:
 ...
 {note}
 ...

Please respond with valid JSON only, no additional text. Output:

2.8. Cross-Chunk Date Extraction (Multi-chunk Context)

Purpose. Continues the same date processing step when notes are chunked; uses the admission date anchor and the previous segment for disambiguation.

```
## Extract dates for ALL entities marked by <KEY> tags in this health record.
    Every entity must have a date entry.
```

```

## Output JSON list of dictionaries with:
- tag: entity order number from KEY tags
- date: [YYYY-MM-DD, YYYY-MM-DD] for start and end (same date if single event;
      null if unknown)
- inferred: [true/false, true/false] indicating whether each boundary was
      inferred

## Key Rules:
- Use admission date as primary anchor
- Consider previous note but prioritize current segment
- Explicit dates -> exact start/end, inferred=false
- "X days/weeks/months/years ago" -> END = admission date, START = admission date
      minus X, inferred=true
- Current symptoms/findings -> admission date for both boundaries, inferred=false
- New treatments/prescriptions -> START = admission date, END = null, inferred=[
      false, true]
- Chronic conditions without timeframe -> [null, null], inferred=[false, false]
- Related events share the main event's window
- Unknown timing -> [null, null], inferred=[false, false]

Input:
...

Previous piece of the record (context):
{prev_note}
Admission date: {adm_date}, Discharge date: {dis_date}
Current note:
{note}
...

```

Please respond with valid JSON only, no additional text. Output:

2.9. Admission and Discharge Date Finder

Purpose. Supports the metadata retention described in Step 1 (chunking) by recovering visit anchors when only coarse dating cues are present.

```

## You will receive an electronic health record for a patient. Your task is
      extract the admission date and the discharge date of the patient.
## Your output should be in JSON Array format: "[admission date, discharge date
      ]". If this does not apply to the record provided, for example, use null to
      fill up the value (such as [null, null]).

```

```

Input:
{note}

```

Please respond with valid JSON only, no additional text. Output:

2.10. Date Normalization (Helper within Step 3)

Purpose. Auxiliary helper used inside the date processing step to normalize relative expressions to absolute start–end ranges using an anchor date; not an additional pipeline stage.

```

## Given an anchor time in the form of YYYY-MM-DD, what is the date range of "{
      date}" when the anchor time is {anchor}?
## Your answer should be given in the form of [YYYY-MM-DD, YYYY-MM-DD]
      representing the start and end time.

```

Please respond with valid JSON only, no additional text. Output:

2.11. JSON Repair (Robustness Helper)

Purpose. Internal robustness helper that auto-corrects malformed JSON from LLM calls; it does not introduce a new pipeline step and only safeguards existing outputs.

```
## I have a JSON file with errors that cannot be parsed. When I tried to parse it
    using `demjson3.decode`, I received specific error messages. Please correct
    the JSON and respond with only the corrected JSON content--do not include
    any explanations, comments, or additional text.
```

```
Error: {error_message}
Input JSON:
{json_content}
```

```
## Please respond with only the corrected JSON. Output:
```

3. JSON Output Specification

The pipeline aggregates module outputs into a consolidated JSON structure before any format conversion. Field names are presented in monospace to distinguish schema tokens from prose. Missing or inapplicable entries are set to null.

Field	Type	Definition
term_index	integer	Position of the entity in the note after ordering.
key	string	Note or encounter identifier provided to the system.
admission_date, discharge_date	string (YYYY-MM-DD)	Visit-level anchors propagated to each entity.
gender, death_date, birth_date, race, ethnicity, zip_code	string	Demographic attributes extracted once per note.
mention	string	Original entity span text.
context	string	Surrounding phrase or sentence for auditability.
mention_start_pos, mention_end_pos	integer	Character offsets within the original note (start inclusive, end exclusive).

Field	Type	Definition
code	string	Ontology identifier concatenated with preferred term as <code>code preferred_term</code> . Derived via SapBERT retrieval over UMLS.
type	string	UMLS semantic type associated with <code>code</code> .
assertion_status	categorical	One of Present, Absent, Historical, Possible, Conditional, Hypothetical, Notassociated.
body_location	string	Anatomical site text.
body_location_code	string	Ontology mapping for <code>body_location</code> using the same concatenation convention as <code>code</code> .
value	string or array	Numeric or qualitative measurement linked to the entity; arrays hold parallel multi-value entries.
unit	string or array	Measurement unit(s) aligned with <code>value</code> .
infer	boolean or array	Indicates whether each unit was inferred rather than explicit.
note	string or array	Comparator for numeric values (greater , lower , equal) or brief qualifier.
freq	string	Medication frequency phrase.
route	string	Medication administration route.
other	string	Additional descriptors not captured elsewhere (for example "fasting", "irregular").
related	object	Directed relations to other entities; keys are target tags and values are relation types.
begin_date, end_date	string (YYYY-MM-DD)	Normalized start and end dates for the entity.
begin_date_inferred, end_date_inferred	boolean	Flags indicating whether each date boundary was inferred.

4. i2b2 Export Schema

Structured outputs are rendered to the i2b2 `OBSERVATION_FACT` layout with lightweight extensions for clinical NLP. Each entity produces one base row plus optional modifier rows when additional attributes are present. Field names follow the canonical i2b2 spelling to preserve compatibility.

4.1. Field Definitions

Field	Type	Definition
patient_num	integer	Pseudonymized patient identifier (provided upstream).

Field	Type	Definition
encounter_num	integer or string	Visit identifier; equals the pipeline key for the note.
birth_date, death_date	date	Demographic dates when present.
sex_cd, race_cd, ethnicity_cd, zip_cd	string	Demographic codes or free-text values from the note.
start_date, end_date	date	Entity time span; defaults to visit anchors when explicit timing is absent.
concept_cd	string	Primary concept code from UMLS mapping (or downstream conversion).
name_char	string	Human-readable mention text.
observation_blob	string	Local context window for audit.
modifier_cd	string	Modifier type; @ for base record, otherwise codes such as DOSE, ROUTE, FREQ, ASSERTION_STATUS, BODY_LOCATION, OTHER_INFO, RELATED.
instance_num	integer	Sequential counter to distinguish multiple modifiers attached to the same entity.
valtype_cd	categorical	N for numeric, T for text.
tval_char	string	Text value or comparator when valtype_cd is text or when expressing greater/lower/equal.
nval_num	numeric	Numeric value when applicable.
valueflag_cd	categorical	Abnormality flags if present (High, Low, Abnormal); otherwise null.
units_cd	string	Measurement unit.
unitflag_cd	boolean (string)	Indicates whether the unit was inferred (true/false).
entity_index	integer	Position of the entity in the original note (copied from term_index).
code_type	string	Coding system label (ICD10CM, LNC, RXNORM, SNOMEDCT_US, or CUI fallback).

4.2. Base and Modifier Records

- **Base record** (modifier_cd = @): carries the primary concept, context, dates, and any value-unit pair directly associated with the entity.
- **Dose, route, frequency** modifiers: emitted when value, route, or freq appear in the JSON output.
- **Assertion status and body location** modifiers: capture clinical status and anatomical site for downstream phenotyping.

- **Other information** modifier: holds descriptive qualifiers not mapped elsewhere.
- **Related** modifier: serializes the directed relationship dictionary to maintain cross-entity links.

5. UMLS Retrieval Procedure

The concept enrichment module aligns cleaned entity mentions to UMLS concepts using SapBERT embeddings (`cambridgelt1/SapBERT-from-PubMedBERT-fulltext`) with FAISS nearest-neighbour search under an inner-product metric on L_2 -normalized vectors (equivalent to cosine similarity). Two independent indices are built: an *all-terms* index over the full deduplicated UMLS term list, and a restricted *body-location* index used for anatomical-site resolution. The index type is selected by corpus size: for term lists larger than 50,000 entries an inverted-file index (`IndexIVFFlat`) with an `IndexFlatIP` coarse quantizer is used, with the number of Voronoi cells set to $nlist = \min(4096, \max(\lfloor N/50 \rfloor, 256))$ and query breadth $nprobe = \min(32, nlist)$; for smaller lists an exact `IndexFlatIP` index is used. The pseudocode below mirrors the deployed implementation.

Algorithm 1 UMLS Concept Retrieval with SapBERT and FAISS

- 1: Load the UMLS term list with concept unique identifiers (CUIs) and semantic types; deduplicate while preserving term–CUI pairs.
 - 2: Encode all terms with the SapBERT encoder; apply L_2 normalization to obtain unit vectors of dimension d .
 - 3: **Index construction** (per index: all-terms, body-location):
 1. If $N > 50,000$: set $nlist = \min(4096, \max(\lfloor N/50 \rfloor, 256))$; build `IndexIVFFlat` with an `IndexFlatIP` quantizer and `METRIC_INNER_PRODUCT`; train on the embeddings (subsampling to at most 10^6 vectors when $N > 10^6$); add all vectors.
 2. Else: build an exact `IndexFlatIP` index and add all vectors.
 3. Attempt GPU construction (`StandardGpuResources`); on GPU failure, retry the same construction on CPU.
 - 4: **Query** (batched): filter out empty or null mentions; encode remaining mentions with the same SapBERT encoder and L_2 normalization; cast to `float32`.
 - 5: For `IndexIVFFlat`, set $nprobe = \min(32, nlist)$; search the appropriate index for the top- k matches ($k = 1$ by default).
 - 6: For each mention, return a JSON object mapping the retrieved CUI to its preferred term and semantic type; attach the retrieved codes to the corresponding entities.
-

Body-location mentions are additionally resolved against the body-location index so that anatomical sites map to anatomy-typed concepts rather than to the nearest concept of any semantic type. The retrieval is deterministic given the encoder, the term list, and the index parameters above; no external network calls are made during concept mapping.

6. Mapping UMLS Concepts to Source Terminologies

CLINES normalizes each extracted mention to a UMLS Concept Unique Identifier (CUI) using the SapBERT + FAISS retrieval procedure of the preceding section. The CUI is the interoperability hub: in the UMLS Metathesaurus a single CUI aggregates synonymous terms contributed by many independently maintained source vocabularies, so a CUI can be deterministically crosswalked to any of those vocabularies through the standard `MRCONSO` concept-atom table without further model inference. The i2b2-style export therefore stores, for every entity, the CUI, its UMLS preferred term, and its semantic type (`MRSTY`); downstream consumers obtain a code in their terminology of choice by joining on the CUI.

Downstream use	Target vocabulary (UMLS source)	Example crosswalk from CUI
Problem lists / diagnoses	SNOMED CT (<code>SNOMEDCT_US</code>)	C0020538 → 38341003 (Hypertensive disorder)
Billing / claims	ICD-10-CM (<code>ICD10CM</code>)	C0020538 → I10 (Essential hypertension)
Medications	RxNorm (<code>RXNORM</code>)	C0025598 → 6809 (Metformin)
Laboratory tests	LOINC (<code>LNC</code>)	C0202194 → 4548-4 (Hemoglobin A1c)
Procedures	CPT / SNOMED CT procedure	C0009378 → 44970 (Appendectomy)

Table 3: Because every entity is anchored to a UMLS CUI, a single normalization step yields codes in arbitrarily many downstream terminologies via standard UMLS crosswalks, without re-running the model. The example crosswalks are illustrative of widely used UMLS source mappings; the specific target codes available for a given CUI depend on the licensed UMLS source vocabularies.

This design decouples extraction from any single coding standard: institutions that require SNOMED CT for clinical documentation, ICD-10-CM for reimbursement, or RxNorm for medication reconciliation all consume the same CUI-anchored output, and the mapping layer is auditable because each CUI carries its preferred term and semantic type alongside the source-vocabulary codes.

7. Detailed Performance Metrics with Bootstrap Confidence Intervals

This section expands Figure 5a of the main text with the full per-field, per-dataset breakdown for the four full-pipeline backbones evaluated under the identical CLINES workflow. For every field we report the point F1 with a bias-corrected 95% bootstrap confidence interval (2,000 resamples over notes; seed 20260525). *Code* F1 requires both correct span detection and correct UMLS

concept assignment; *Assertion*, *Value*, and *Unit* are conditional on a correctly matched mention; *Begin/End date* F1 reflect temporal-boundary agreement after relative-date normalization. The mention-level breakdown and single-prompt / deep-learning baselines appear in Figure 5d.

Backbone	Dataset	Code	Assertion	Value	Unit	Begin date	End date
o3-mini-medium	4CE	0.874 (.849-.896)	0.884 (.857-.909)	0.815 (.748-.867)	0.771 (.710-.826)	0.555 (.495-.624)	0.560 (.498-.637)
	CORAL-B	0.814 (.787-.838)	0.840 (.803-.871)	0.801 (.711-.858)	0.752 (.675-.810)	0.683 (.614-.756)	0.690 (.622-.764)
	CORAL-P	0.848 (.834-.865)	0.873 (.857-.887)	0.905 (.877-.932)	0.895 (.863-.926)	0.708 (.630-.783)	0.722 (.638-.797)
GPT-4o	4CE	0.781 (.759-.801)	0.842 (.825-.858)	0.771 (.715-.822)	0.764 (.698-.823)	0.780 (.627-.867)	0.729 (.443-.847)
	CORAL-B	0.701 (.667-.731)	0.761 (.714-.799)	0.617 (.546-.683)	0.545 (.476-.612)	0.734 (.636-.820)	0.735 (.634-.821)
	CORAL-P	0.781 (.751-.810)	0.852 (.831-.872)	0.857 (.808-.900)	0.761 (.687-.840)	0.780 (.706-.843)	0.772 (.698-.836)
Llama-3.1-405B	4CE	0.665 (.641-.690)	0.738 (.715-.760)	0.711 (.660-.761)	0.655 (.588-.713)	0.706 (.573-.767)	0.608 (.208-.715)
	CORAL-B	0.608 (.577-.636)	0.706 (.672-.733)	0.667 (.590-.734)	0.735 (.671-.791)	0.653 (.600-.714)	0.630 (.542-.713)
	CORAL-P	0.655 (.624-.686)	0.746 (.722-.770)	0.776 (.705-.836)	0.720 (.649-.782)	0.629 (.540-.706)	0.635 (.550-.707)

Table 4: Per-field F1 with 95% bootstrap confidence intervals for the four full-pipeline backbones across the three expert-annotated datasets (4CE; CORAL breast cancer, CORAL-B; CORAL pancreatic cancer, CORAL-P). CORAL-P value/unit fields exceed code F1 because numeric laboratory and pathology measurements are densely and explicitly reported in oncology synoptic notes. Date-field F1 is uniformly lower, reflecting the difficulty of resolving relative and implicit temporal expressions.

8. Inter-Annotator Agreement: Per-Note Detail

This section provides the per-note inter-annotator agreement (IAA) results underlying Figure 4a and the headline agreement statistic in the main text. For the cross-annotation study, three notes per dataset (4CE; CORAL) were independently re-annotated by a second annotator, blinded to the original assignment. We report agreement before (*pre*) and after (*post*) applying the five pre-specified structural-disambiguation guidelines (Figure 4b), using three complementary statistics: mention-level F1, Cohen’s κ , and prevalence-and-bias-adjusted κ (PABAK). F1-based agreement is our primary metric because the keep/discard class distribution is highly skewed ($\approx 75\%$ keep), conditions under which κ is depressed by the base rate (“Cohen’s paradox”); we therefore report all three for transparency.

Dataset	Note	<i>n</i> rows	F1 (pre→post)	κ (pre→post)	PABAK (pre→post)	Raw % (pre→post)
4CE	report03	334	.891→.901	.470→.636	.653→.711	82.6→85.5
4CE	report04	299	.812→.850	.251→.583	.431→.612	71.6→80.6
4CE	KUMC_7	371	.668→.705	.295→.503	.224→.485	61.2→74.2
CORAL	pdac_17	443	.906→.905	.743→.769	.761→.777	88.0→88.8
CORAL	pdac_7	554	.850→.859	.383→.570	.545→.621	77.3→81.0
CORAL	pdac_14	656	.787→.775	.380→.510	.439→.510	72.0→75.5
<i>Pooled</i>		2,657	0.837	0.597	0.609	80.5

Table 5: Per-note IAA on the cross-annotation subset (three notes per dataset; 4CE and CORAL). The pre-specified structural-disambiguation guidelines improve agreement on every metric for almost all notes. The pooled mention-level F1 reported in the main text is 0.837 ($\kappa = 0.597$, PABAK = 0.609, raw agreement 80.5%), within the substantial-agreement range reported for clinical named-entity IAA (cf. i2b2 2010 challenge $\kappa = 0.43$ –0.60 for entity-boundary tasks). κ is uniformly lower than F1 because, under the highly skewed keep/discard distribution, it is depressed by the base rate; PABAK and the pre→post improvement confirm that the reconciliation guidelines resolve guideline-ambiguity divergence rather than substantive clinical disagreement.

9. Hallucination Taxonomy and Case Studies

To characterize the nature of CLINES’ apparent false positives (FPs) against the gold standard, we drew a stratified sample of candidate FP predictions and had each manually adjudicated into a seven-class taxonomy by a reviewer with access to the full source note. Because the sample was stratified by an automatic rule-based pre-classifier, we extrapolate the per-stratum judge distribution to the full FP population ($N = 7,171$) by post-stratified Horvitz–Thompson estimation: within each rule stratum the judged category proportions are scaled to that stratum’s population size and summed across strata. The seven classes distinguish genuine model errors (fabricated entity, span-boundary error, wrong assertion, wrong code, wrong value, wrong date) from “not an error”, where the prediction is in fact defensible and the discrepancy reflects a gap or inconsistency in the reference annotation rather than a model hallucination.

Category	Extrapolated count	% of candidate FPs
Not an error (annotation gap)	3,895	54.31
Wrong code	1,199	16.72
Wrong assertion	621	8.67
Wrong value	596	8.31
Fabricated entity	388	5.41
Span-boundary error	387	5.39
Wrong date	85	1.19
True hallucination (6 categories)	3,276	45.69
Total candidate FPs	7,171	100.0

Table 6: Seven-class hallucination taxonomy over the full candidate-FP population, extrapolated from $n = 700$ manually judged predictions via post-stratified Horvitz–Thompson estimation. The single largest category is “Not an error”: a majority (54.3%) of apparent FPs are defensible predictions that the reference annotation omitted or recorded inconsistently. No individual *true*-hallucination category exceeds 17% of FPs, and outright “Fabricated entity” errors account for only 5.4%.

9.1. Representative case studies

The following examples are drawn verbatim (mentions and adjudication rationales) from the manual-judging study and illustrate each major category.

Not an error (annotation gap).

CLINES extracted “*RLS*” and normalized it to *restless legs syndrome* (C0035258). The abbreviation and its expansion are explicitly present in the note’s problem list, but the rule-based reference annotation did not capture it. The prediction is correct; the discrepancy is a gap in the gold standard.

Wrong code.

For “*HR+/HER2- breast cancers*” CLINES assigned C3539878 (*ER-negative, PR-negative, HER2-negative breast cancer*). The surrounding text describes a hormone-receptor-*positive*, HER2-negative tumor, so the assigned concept inverts the biomarker status—a genuine normalization error.

Wrong assertion.

“*Penicillins*” (C0030842) was marked Present, but the note states the patient’s penicillin-allergy status is uncertain (“not sure if she is allergic”), which the schema treats as Conditional. The concept is correct; the assertion class is wrong.

Span-boundary error.

CLINES extracted “*TYLENOL*” where the gold span is “*acetaminophen (TYLENOL)*”. The normalized concept and code (C0699142) are identical and correct; only the surface span is truncated.

Fabricated entity.

“*Age of Onset*” was extracted (and coded to C3175531) from a *column header* in a family-history table rather than from any patient-specific clinical

statement—an entity that does not correspond to a real clinical concept in context.

These case studies motivate the main-text argument that raw precision against a single-pass reference annotation systematically understates CLINES’ true reliability, because the dominant “error” mode is recovery of clinically valid mentions the reference omitted, not fabrication.

10. Output Consistency Across Repeated Runs

Large-language-model outputs can vary across repeated invocations even at low decoding temperature, owing to non-deterministic kernel scheduling and floating-point non-associativity on accelerator hardware. Because CLINES is intended for chart-review-like tasks where reproducibility matters, we assess run-to-run stability of the end-to-end pipeline with a pre-specified protocol.

10.1. Protocol

We select a fixed subset of 20 notes (balanced across the 4CE and CORAL datasets) and run the complete CLINES pipeline $R = 5$ times independently on each note with identical inputs and configuration (gpt-4o-mini backbone; temperature as deployed; distinct random seeds where exposed by the API). Mention identity is defined *semantically*, not by exact span string: two extracted mentions are considered equivalent if they share a non-empty UMLS code or share a canonical surface form (lowercased, punctuation-stripped, whitespace-collapsed). This absorbs span-boundary jitter and surface synonyms that the pipeline mapped to the same concept, both of which exact-span matching would otherwise count as disagreement. For each note we compare the five extracted i2b2 record sets and quantify stability with:

- *Pairwise mention-set Jaccard* (over semantic equivalence-class ids) and *code-set Jaccard* across the $\binom{5}{2} = 10$ run pairs, summarizing the overlap of extracted mentions and assigned UMLS codes;
- *Per-field majority-vote stability*: for every equivalence class that appears in at least two runs, the fraction of those runs whose value for a given field (assertion status, value, unit, begin date, end date) equals the modal value across runs; averaged over all such (class, field) cells. When a class fires multiple times within one run, that run’s representative value for the cell is the run’s own modal value for the class.

This protocol mirrors the output-consistency assessment recommended by Ntinopoulos et al. (BMJ Health & Care Informatics 2025) for clinical LLM evaluations, which the present study did not originally include.

10.2. Results

All 20 notes produced complete outputs in all five runs (no run failures; no partial outputs). Across the $\binom{5}{2} = 10$ run pairs per note, the mean semantic mention-set Jaccard was 0.752 (per-note range 0.664–0.861) and the mean code-set Jaccard was 0.622 (range 0.498–0.718); the gap of ≈ 0.13 between mention-set and code-set Jaccard reflects mentions for which the pipeline assigned no UMLS code (rather than disagreement at the concept level). Approximately three quarters of extracted mentions and roughly 62% of UMLS codes are therefore shared across any two runs on the same note, consistent with stochastic LLM decoding at non-zero temperature.

For mentions that *are* recovered across runs, the structured fields associated with each mention are highly stable: averaging over all 27,605 (class, field) cells with at least two contributing runs, the modal field value was matched in 89.9% of contributing runs on average (Table 7). Field-level stability was highest for unit (94.9%), end date (94.2%), and begin date (91.3%), and slightly lower for assertion status (84.9%) and numeric value (84.3%). Together these two views indicate that, although the LLM does not deterministically reproduce the full extracted set on every run, the per-field content of each extracted mention is largely run-invariant; in chart-review use, an analyst aggregating multiple runs (e.g., via majority vote over $R \geq 3$ runs) recovers a stable consensus output.

Table 7: Per-field majority-vote stability across $R = 5$ independent runs on the 20-note subset (mean fraction of contributing runs equal to the modal value; n_{cells} is the number of (mention-class, field) cells with ≥ 2 contributing runs, using semantic mention identity).

Field	Mean modal agreement	Per-cell range	n_{cells}
Assertion status	0.849	[0.250, 1.000]	5,521
Numeric value	0.843	[0.200, 1.000]	5,521
Unit	0.949	[0.200, 1.000]	5,521
Begin date	0.913	[0.200, 1.000]	5,521
End date	0.942	[0.200, 1.000]	5,521
All fields pooled	0.899	[0.200, 1.000]	27,605

11. Reporting Checklists: MI-CLAIM and TRIPOD+AI

Because CLINES is a clinical artificial-intelligence system rather than an observational epidemiological study, we report it against AI-specific reporting standards—the Minimum Information about Clinical Artificial Intelligence Modeling (MI-CLAIM) checklist and the TRIPOD+AI statement—in place of the STROBE checklist used for observational studies. The tables below map each checklist’s principal items to where they are addressed in the manuscript and supplement.

MI-CLAIM item	Where addressed
1. Study design, clinical problem, cohort/data source, ground truth	Methods (study design, datasets, annotation protocol); Ethics Approval
2. Data preparation, partitioning, and pre-processing	Methods (chunking and pre-processing); Supplementary prompt suite
3. Model description and optimization (architecture, training-free use of LLMs)	Methods (pipeline stages); Supplementary prompt suite and retrieval procedure
4. Model performance, evaluation metrics, uncertainty	Results; Figures 3–5; Supplementary detailed-metrics and IAA tables
5. Model examination (error analysis, ablation, robustness)	Results (ablation, hallucination analysis, note-length); Figures 4–5; Supplementary hallucination and consistency sections
6. Reproducibility (code, prompts, models, data access tier)	Data Sharing (MIT-licensed reference implementation); Supplementary prompt suite, schemas, and retrieval pseudocode

Table 8: MI-CLAIM (Norgeot et al., Nat Med 2020) item-to-location mapping. CLINES is used in a training-free configuration, so MI-CLAIM optimization items pertaining to model training are addressed by the prompt-suite and retrieval specifications rather than by a training procedure.

TRIPOD+AI item (grouped)	Where addressed
Title and structured abstract	Title; Abstract (Objective / Methods and Analysis / Results / Conclusion)
Background and objectives	Introduction
Data sources, settings, participants	Methods (datasets); Ethics Approval; Data Sharing
Predictors/inputs, outcome/target, ground-truth definition	Methods (task definition, i2b2 schema, annotation)
Sample size and missing data	Methods; Results (per-dataset note and mention counts)
AI model and analytical methods	Methods (pipeline); Supplementary prompt suite and retrieval procedure
Model performance and uncertainty quantification	Results; Figures 3–5; Supplementary detailed-metrics (bootstrap 95% CIs)
Model evaluation, comparison with baselines, ablation	Results; Figure 5; Supplementary tables
Limitations and generalizability	Discussion (Limitations; Dataset-level variation)
Funding, conflicts, data and code availability	Funding; Declaration of Interests; Data Sharing

Table 9: TRIPOD+AI (Collins et al., BMJ 2024) item-to-location mapping, grouped by checklist domain. Items specific to incremental-value and calibration analyses for risk-prediction models are not applicable to an information-extraction system and are reported as not applicable.